Real Variables II

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Chapter 5

Differentiation review

Theorem 5.1 (Rolle's Theorem). Let $a < b$ be real numbers, and let $g : [a, b] \to \mathbb{R}$ be a continuous function which is differentiable on (a, b). Suppose that $g(a) = g(b)$. Then there exists an $x \in (a, b)$ such that $g'(x) = 0$.

Theorem 5.2 (Mean Value Theorem). If f is a real continuous function on $[a, b]$ which is differentiable in (a, b), then there is a point $x \in (a, b)$ at which

$$f(b) - f(a) = (b - a)f'(x).$$

Theorem 5.3 (Taylor's theorem with error). Let $f : [a, b] \to \mathbb{R}$, let $n \geq 0$ be an integer and suppose that f is $(n + 1)$ times differentiable on $[a, b]$. Let x_0 and x be points in $[a, b]$ with $x_0 \neq x$. Then there exists some point c between x_0 and x such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n + 1)!}(x - x_0)^{n+1}. \tag{5.1}$$

To warm up to the proof, we can consider the case where $n = 0$. From (5.1) we then get

$$f(x) = f(x_0) + f'(c)(x - x_0).$$

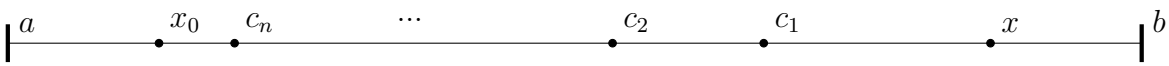
Observe that that we can reorder this equation to match MVT, so we know it to be true. This gives us the intuition the Taylor's theorem is a higher order generalization of the MVT.

Proof. Define $A \in \mathbb{R}$ by $f(x) - P_n(x) = \dfrac{A}{(n + 1)!}(x - x_0)^{n+1}$. We want to show that there is c between x and x_0 such that $f^{n+1}(c) = A$.
Now, define $g(t) = f(t) - P_n(t) - \dfrac{A}{(n + 1)!}(t - x_0)^{n+1}$.
Observe that $g(x_0) = f(x_0) - P_n(x_0) = f(x_0) - f(x_0) = 0$ and $g(x) = f(x) - P_n(x) - \dfrac{A}{(n + 1)!}(x - x_0)^{n+1} = 0$ by the definition of A.
Then, for $j = 0, ..., n$, $g^{(j)}(x_0) = f^{(j)}(x_0) - P_n^{(j)}(x - x_0) - \dfrac{\mathrm{d}^j}{\mathrm{d}t^j}\dfrac{A}{(n + 1)!}(t - x_0)^{n+1}\Big|_{t=x_0}$ and $g^{(n+1)}(t) = f^{(n+1)}(t) - 0 - A$.

Now, we can shift our goal to show that there exists c between x and x_0 such that

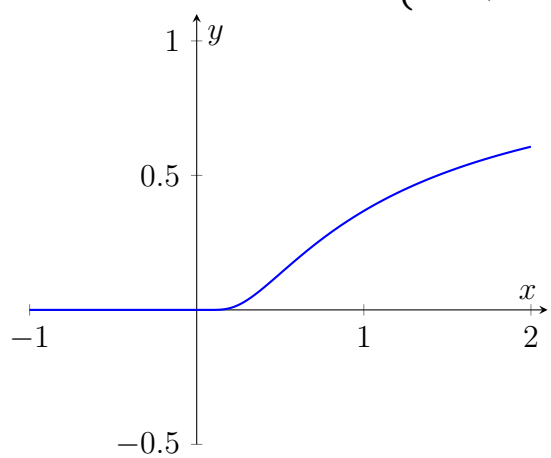
$$g^{(n+1)}(c) = 0. \tag{5.2}$$

We know $g(x) = 0 = g(x_0)$ so by theorem 5.1, there is c_1 between x and x_0 such that $g'(c_1) = 0$. Then $g'(x_0) = 0 = g'(c)$ so there is c_2 between c_1 and x_0 such that $g''(c_2) = 0$. We can repeat this process to obtain the set $\{c_1, c_2, ..., c_n\}$ where $g^{(i)}(c_i) = 0$. Then, we know $g^{(n)}(x_0) = 0 = g^{(n)}(c_n)$ so there is c_{n+1} between x_0 and c_n such that $g^{(n+1)}(c_{n+1}) = 0$.
Let $c = c_{n+1}$. Then c satisfies (5.2) as required.



$\square$

Example. Consider $f(x) = \begin{cases} 0, x \leq 0 \\ e^{-1/x}, x > 0 \end{cases}$



Then for all $x > 0, k \in \mathbb{N}$

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k} f(x) = Q(x)e^{-1/x}$$

where $Q(x)$ is some quotient.
We could show all the derivatives are continuous at 0 by computing the limits on each side, with limit 0. So, this function has Taylor polynomial about 0 of $P_n(t) = 0$ for all n.
This shows that Taylor polynomials do no encode all the information necessary to extrapolate what function is encoded.

Chapter 6

The Riemann-Stieltjes Integral

6.1 Definition and Existence of the Integral

Definition 6.1. A **partition** of $[a, b]$ is a finite set $P = \{x_0, x_1, ..., x_n\}$ with $a = x_0 < x_1 < ... < x_n = b$.
For each $i = 1, ..., n$ let $\Delta x_i = x_i - x_{i-1}$.
Now suppose $f : [a, b] \to \mathbb{R}$ is a bounded function, we denote

$$M_i = \sup \{f(x) : x \in [x_{i-1}, x_i]\} \text{ and } m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}.$$

as well as

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i \text{ and } L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i$$

and finally

$$\text{The upper Riemann integral: } \overline{\int_a^b} f \mathrm{d}x = \inf U(P, f)$$

$$\text{The lower Riemann integral: } \underline{\int_a^b} f \mathrm{d}x = \sup L(P, f)$$

where the inf and sup are taken over all partitions P of $[a, b]$. If the upper and lower integrals agree with each-other we say f is Riemann integrable and denote the common value by

$$\int_a^b f \mathrm{d}x.$$

We say $f \in \mathcal{R}[a, b]$, where $\mathcal{R}[a, b]$ is the set of Riemann integrable functions on $[a, b]$.

Example. Let $[a, b] = [0, 1]$ and $f(x) = x$.
If $P = \{x_0, ..., x_n\}$ is a partition, $M_i = x_i$ and $m_i = x_{i-1}$.
Consider $P = \{0, 1/n, 2/n, \ldots, 1\}$, then,

$$U(P, f) = \sum_{i=1}^{n} \frac{i}{n} \frac{1}{n} = \frac{1}{n^2} \sum_{i=1}^{n} i = \frac{1}{n^2} \frac{1}{2} n(n+1) = \frac{1}{2} + \frac{1}{2n}.$$

In particular,

$$\int_0^1 x\mathrm{d}x \leq \inf\left\{\frac{1}{2}+\frac{1}{2n}|n\in\mathbb{N}\right\}.$$

And

$$L(P,f)=\sum_{i=1}^{n}\frac{i-1}{n}\frac{1}{n}=\frac{1}{n^2}\sum_{i=1}^{n}i-1=\frac{1}{n^2}\frac{1}{2}n(n-1)=\frac{1}{2}-\frac{1}{2n}.$$

In particular,

$$\int_0^1 x\mathrm{d}x \geq \sup\left\{\frac{1}{2}-\frac{1}{2n}|n\in\mathbb{N}\right\}.$$

We can now ask ourselves: is this enough to conclude that f is in fact Riemann integrable on $[0,1]$?
Well... not quite yet, we need a bit more work for that.

Definition 6.2. Let $\alpha:[a,b]\to\mathbb{R}$ be (weakly) monotone increasing, let $P=\{x_0,x_1,...,x_n\}$ be a partition of $[a,b]$.
For $i=1,\ldots,n$ let $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})(\geq 0)$.
For $f:[a,b]\to\mathbb{R}$ bounded, we define

$$U(P,f,\alpha)=\sum_{i=1}^{n}M_i\Delta\alpha_i \text{ and } L(P,f,\alpha)=\sum_{i=1}^{n}m_i\Delta\alpha_i$$

The upper Riemann-Stieltjes integral: $\overline{\int_a^b} f\mathrm{d}\alpha=\inf U(P,f,\alpha)$

The lower Riemann-Stieltjes integral: $\underline{\int_a^b} f\mathrm{d}\alpha=\sup L(P,f,\alpha)$

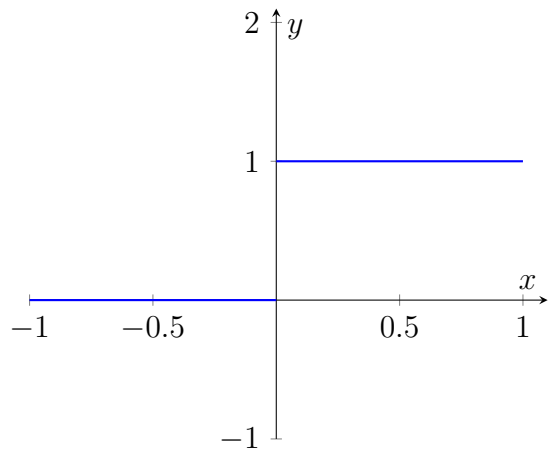
If the upper and lower Riemann-Stieltjes integrals agree, we denote the common value by

$$\int_a^b f\mathrm{d}\alpha$$

and we say $f\in\mathcal{R}_\alpha[a,b]$.

Note. It is not a requirement that α is continuous on $[a,b]$.

Consider $\alpha=\begin{cases}0, & x\leq 0\\ 1, & x>0\end{cases}$



What does $\int_{-1}^{1} f \mathrm{d}\alpha$ look like ? This will be equal to $f(0)$, like integrating $f(x)\delta_0$ where δ_0 is the Dirac-delta function.

Definition 6.3. Let P and P^* be partitions of $[a, b]$. We say P^* is a **refinement** of P if $P \subset P^*$.
If P_1 and P_2 are partitions, their **common refinement** is the partition $P_1 \cup P_2$.

Theorem 6.4. Let P^* be a refinement of P. Then

$$L(P, f, \alpha) \leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P, f, \alpha).$$

Proof. For a non-empty set its infimum is smaller or equal to its supremum, so the middle inequality holds.
Since partitions are finite, it suffices to prove the inequality when P^* contains one additional point x^*.
Let $x_i < x^* < x_{i+1}$. We can then compare $L(P, f, \alpha)$ and $L(P^*, f, \alpha)$.
Put

$$w_1 = \inf \{f(x) | x \in [x_i, x^*]\}$$
$$w_2 = \inf \{f(x) | x \in [x^*, x_{i+1}]\}$$

$$L(P, f, \alpha) = \sum_{j=1}^{n} m_j \Delta\alpha_j$$
$$L(P^*, f, \alpha) = \sum_{j=1}^{i} m_j \Delta\alpha_j + w_1(\alpha(x^*) - \alpha(x_i)) + w_2(\alpha(x_{i+1}) - \alpha(x^*)) + \sum_{j=i+2}^{n} m_j \Delta\alpha_j$$

Then

$$\begin{aligned} L(P^*, f, \alpha) - L(P, f, \alpha) &= w_1(\alpha(x^*) - \alpha(x_i)) + w_2(\alpha(x_{i+1}) - \alpha(x^*)) - m_{i+1}\Delta\alpha_{i+1} \\ &\leq m_{i+1}(\alpha(x^*) - \alpha(x_i)) + m_{i+1}(\alpha(x_{i+1}) - \alpha(x^*)) - m_{i+1}\Delta\alpha_{i+1} \\ &= 0 \end{aligned}$$

The proofs for the upper sum is similar. □

Theorem 6.5. Let $f : [a, b] \to \mathbb{R}$ be bounded and $\alpha : [a, b] \to \mathbb{R}$ be monotonically increasing then

$$\underline{\int_b^a} f \mathrm{d}\alpha \leq \overline{\int_b^a} f \mathrm{d}\alpha.$$

Proof. Let P_1 and P_2 be partitions of $[a, b]$, then by Theorem (6.4) we know

$$L(P_1, f, \alpha) \leq U(P_2, f, \alpha)$$

hence

$$\underline{\int_b^a} f \mathrm{d}\alpha = \sup_{P_1} L(P_2, f, \alpha) \leq U(P_2, f, \alpha).$$

Since this is true for every P_2,

$$\underline{\int_b^a} f \mathrm{d}\alpha \leq \inf_{P_2} U(P_2, f, \alpha) = \overline{\int_b^a} f \mathrm{d}\alpha.$$

□

This theorem was the missing piece to show $\int_0^1 x \mathrm{d}x = \frac{1}{2}$.

Theorem 6.6. Let $f : [a, b] \to \mathbb{R}$ be bounded and $\alpha : [a, b] \to \mathbb{R}$ be monotonically increasing. Then

$$f \in \mathcal{R}_\alpha[a, b] \Leftrightarrow \forall \varepsilon, \exists P \text{ such that } U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Proof. ($\Rightarrow$) By hypothesis,

$$\sup_P L(P, f, \alpha) = \int_b^a f \mathrm{d}\alpha = \inf_P U(P, f, \alpha).$$

Let $\varepsilon > 0$ then there exists a partition P_1 such that $L(P_1, f, \alpha) > \int_b^a f \mathrm{d}\alpha - \frac{\varepsilon}{2}$ and there exists a partition P_2 such that $U(P_2, f, \alpha) < \int_b^a f \mathrm{d}\alpha - \frac{\varepsilon}{2}$.
Let $P = P_1 \cup P_2$, then by Theorem (6.4)

$$L(P_1, f, \alpha) \leq L(P, f, \alpha) \leq U(P, f, \alpha) \leq U(P_1, f, \alpha).$$

Hence $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$.
($\Leftarrow$) This follows from the definition. □

Theorem 6.8. Let $\alpha : [a, b] \to \mathbb{R}$ be weakly monotone increasing and f be continuous, then $f \in \mathcal{R}_\alpha[a, b]$.

Proof. f is continuous and $[a, b]$ is compact hence f is uniformly continuous.
Hence for all $\varepsilon_1 > 0$, there is a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon_1$ for all x, y with $|x - y| < \delta$.
Thus, if P is a partition with $\Delta x_i < \delta/2$ for all x_i then

$$M_i - m_i < \varepsilon_1$$

for all i. Hence

$$U(P, f, \alpha) - L(P, f, \alpha) \leq \sum_{i=0}^{n} \varepsilon_1 \Delta\alpha_i = \varepsilon_1(\alpha(b) - \alpha(a)).$$

Given $\varepsilon > 0$, select ε_1 sufficiently small such that

$$\varepsilon_1(\alpha(b) - \alpha(a)) < \varepsilon.$$

Choose P as above for the corresponding ε_1.
Then by Theorem (6.6), $f \in \mathcal{R}_\alpha[a, b]$. □

Theorem 6.9. Let $f : [a, b] \to \mathbb{R}$ be monotone and $\alpha : [a, b] \to \mathbb{R}$ be weakly increasing and continuous then $f \in \mathcal{R}_\alpha[a, b]$.

Proof. Let $n \in \mathbb{N}$. By the intermediate value theorem, there is a partition P such that

$$\Delta\alpha_i = \frac{\alpha(b) - \alpha(a)}{n}.$$

Suppose f is monotone increasing, then $M_i = f(x_i)$ and $m_i = f(x_{i-1})$.
Then

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^{n} (M_i - m_i) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^{n} (f(x_i) - f(x_{i-1}) \\ &= \frac{\alpha(b) - \alpha(a)}{n} (f(b) - f(a)). \end{aligned}$$

Given $\varepsilon > 0$, select $n \in \mathbb{N}$ such that

$$\left| \frac{1}{n}(\alpha(b) - \alpha(a))(f(b) - f(a)) \right| < \varepsilon.$$

For such a partition P,

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

By Theorem (6.6), $f \in \mathcal{R}_\alpha[a, b]$.
If f is decreasing an analogous proofs holds. □

Theorem 6.10. Let $f : [a, b] \to \mathbb{R}$ be bounded and continuous at finitely many points. Let $\alpha : [a, b] \to \mathbb{R}$ be monotone increasing and continuous at every point at which f is discontinuous. Then $f \in \mathcal{R}_\alpha[a, b]$.

Proof. Let $E = \{e_1, e_2, \ldots, e_k\}$ be the set of points where f is discontinuous, $\varepsilon_1 > 0$ and

$$N = \sup_{x \in [a,b]} |f| .$$

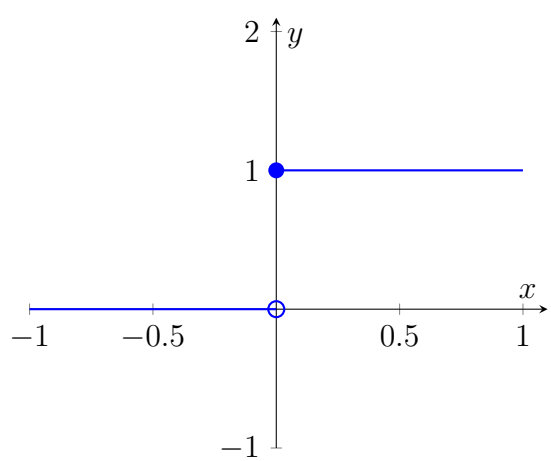
Since α is continuous at each e_i, we can choose $u_i < e_i < v_i$ s.t $0 < \alpha(v_i) - \alpha(u_i) < e_1$, including equality of $e_i = a$ or $e_i = b$.
Let $k = [a, b] \setminus \cup_{i=1}^{k} (u_i, v_i)$. Then k is compact and f is continuous on k so f is uniformly continuous on k, i.e. there exists $\delta > 0$ such that for all x, y in k, if $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon_1$.
Let $\{y_i\} \subset k$ be a set of points so that for every $x \in k$, there is an index i such that $y_i < x < y_{i+1}$ and $0 < y_{i+1} - y_i < \delta$.
Let $P = \{u_i, v_i\}_{i=1}^{k} \cup \{y_i\} \cup \{a, b\}$ (Reordering as necessary to put things in increasing order). Then

$$\begin{aligned} 0 < U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i \\ &\leq k(2N)\varepsilon_1 + \varepsilon_1(\alpha(b) - \alpha(a)) \end{aligned}$$

Given $\varepsilon > 0$, we can choose ε_1 appropriately such that $k(2N)\varepsilon_1 + \varepsilon_1(\alpha(b) - \alpha(a)) < \varepsilon$, and use the corresponding partition P. $\square$

Example. What if f and α are discontinuous at common points? Is f always in $\mathcal{R}_\alpha[a, b]$. Sometimes ? Never ?
Consider the following pair of functions:

$$\alpha(x) = f(x) = \begin{cases} 0, x < 0 \\ 1, x \geq 0 \end{cases}$$



Then if $0 \notin P$ then $\sum_{i=1}^{n}(M_i - m_i)\Delta\alpha_i = 1$.
If $0 \in P$, there is some interval $[s, 0] = [x_{i-1}, x_i]$. Then $M_i = 1$, $m_i = 0$ and $\Delta\alpha_i = 1$, so $\sum_{i=1}^{n}(M_i - m_i)\Delta\alpha_i = 1$. So $f \notin \mathcal{R}_\alpha[-1, 1]$.
However with

$$\alpha(x) = \begin{cases} 0, x \leq 0 \\ 1, x > 0 \end{cases},$$

when $0 \in P$, then $U(P, f, \alpha) - L(P, f, \alpha) = 0$. Furthermore $\int_{-1}^{1} f \mathrm{d}\alpha = 0$.
So the answer is sometimes.

Theorem 6.11. Let $f : [a, b] \to \mathbb{R}$ be bounded and $\alpha : [a, b] \to \mathbb{R}$ monotone increasing. Suppose $f \in \mathcal{R}_\alpha[a, b]$, $m \leq f(x) \leq M$ for all $x \in [a, b]$ and $\phi : [m, M] \to \mathbb{R}$ be continuous. Then $\phi \circ f \in \mathcal{R}_\alpha[a, b]$.

Proof. See Rudin page 127. □

6.2 Properties of the Integral

Theorem 6.12. Let $\alpha : [a, b] \to \mathbb{R}$ be monotone increasing, $f_1, f_2, f_3 : [a, b] \to \mathbb{R}, \in \mathcal{R}_\alpha[a, b]$.

(a) **Linearity.**

$$f_1 + f_2 \in \mathcal{R}_\alpha[a, b] \text{ and } \int_a^b (f_1 + f_2)\mathrm{d}\alpha = \int_a^b f_1 \mathrm{d}\alpha + \int_a^b f_2 \mathrm{d}\alpha$$

and for all $c \in \mathbb{R}$,

$$cf \in \mathcal{R}_\alpha[a, b] \text{ and } \int_a^b cf \mathrm{d}\alpha = c\int_a^b f \mathrm{d}\alpha$$

(b) **Non-negativity.** If $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f \mathrm{d}\alpha \geq 0$$

.

And if $f_1(x) \leq f_2(x)$ for all $x \in [a, b]$, then

$$\int_a^b f_1 \mathrm{d}\alpha \leq \int_a^b f_2 \mathrm{d}\alpha.$$

(c) For $c \in (a, b)$, then f is in$\mathcal{R}_\alpha[a, c]$ and $\mathcal{R}_\alpha[c, b]$. Furthermore

$$\int_a^b f \mathrm{d}\alpha = \int_a^c f \mathrm{d}\alpha + \int_c^b f \mathrm{d}\alpha$$

(d) **Boundedness.** If $|f(x)| \leq M$ then

$$\left| \int_a^b f \mathrm{d}\alpha \right| \leq M(\alpha(b) - \alpha(a)).$$

(e) Let $\alpha_1, \alpha_2 : [a, b] \to \mathbb{R}$ be monotone increasing and f in $\mathcal{R}_{\alpha_1}[a, b]$ and $\mathcal{R}_{\alpha_2}[a, b]$. Then $f \in \mathcal{R}_{\alpha_1+\alpha_2}[a, b]$ and

$$\int_a^b f \mathrm{d}(\alpha_1 + \alpha_2) = \int_a^b f \mathrm{d}\alpha_1 + \int_a^b f \mathrm{d}\alpha_2.$$

If $c \in \mathbb{R}$, then $f \in \mathcal{R}_{c\alpha}[a, b]$ and

$$\int_a^b f \mathrm{d}(c\alpha) = c \int_a^b f \mathrm{d}\alpha.$$

Proof. See Rudin page 128. □

Remark. Recall $C([a, b])$ is the space of continuous functions $f : [a, b] \to \mathbb{R}$ with

$$||f||_{C([a,b])} = \sup_{x \in [a,b]} |f(x)|$$

and

$$\rho(f, g) = ||f - g||_{C([a,b])}.$$

The space $(C([a, b]), ||\cdot||_{C([a,b])}$ is in fact a normed vector space.
Property (a) says that if $\alpha : [a, b] \to \mathbb{R}$ is monotone increasing then the function

$$T(f) = \int_a^b f \mathrm{d}\alpha$$

is a linear function from the vector space $C([a, b])$ to $\mathbb{R}$. Property (d) says T is bounded and property (b) says T is non-negative.
In functional analysis (Math 421) and more generally in physics, we want to study linear functions whose domain is $C([a, b])$ (or more general) and whose co-domain is $\mathbb{R}$ or more often $\mathbb{C}$. Functions of this type are called "linear operators" or "linear functions" and are in essence infinite dimension matrices.

Theorem (Riesz Representation Theorem Version 1). Let $T : C([a, b]) \to \mathbb{R}$ be linear, bounded and non-negative. Then there exists a unique $\alpha : [a, b] \to \mathbb{R}$, monotone increasing, such that

$$Tf = \int_a^b \mathrm{d}\alpha.$$

Theorem (Riesz Representation Theorem Version 2). Let T be a real valued linear functional[1] on $C([a, b])$. Then there exist $\alpha, \beta : [a, b] \to \mathbb{R}$, monotone increasing such that

$$Tf = \int_a^b f \mathrm{d}\alpha - \int_a^b f \mathrm{d}\beta.$$

[1] $T : C([a, b]) \to \mathbb{R}$ is linear and bounded.

Theorem 6.13. Let $f, g \in \mathcal{R}_\alpha[a,b]$, then

(a) $fg \in \mathcal{R}_\alpha[a,b]$.

(b) $|f| \in \mathcal{R}_\alpha[a,b]$ and $\left|\int_a^b f \mathrm{d}\alpha\right| \leq \int_a^b |f| \,\mathrm{d}\alpha$.

Proof. Let $f, g \in \mathcal{R}_\alpha[a,b]$.

(a) By Theorem (6.12) a, $f+g$ and $f-g \in \mathcal{R}_\alpha[a,b]$ and by Theorem (6.11) (with $\phi(x) = x^2$), $(f+g)^2$ and $(f-g)^2 \in \mathcal{R}_\alpha[a,b]$.

So again by Theorem (6.12) $(f+g)^2 - (f-g)^2 = 4fg^2 \in \mathcal{R}_\alpha[a,b]$ and (with $c = 1/4$), $fg \in \mathcal{R}_\alpha[a,b]$.

(b) By Theorem (6.11) (with $\phi(x) = |x|$), $|f| \in \mathcal{R}_\alpha[a,b]$. Let $c = \operatorname{sign}(\int_a^b f \mathrm{d}\alpha)$, then

$$\left|\int_a^b f \mathrm{d}\alpha\right| = c\int_a^b f \mathrm{d}\alpha = \int_a^b cf \mathrm{d}\alpha \leq \int_a^b |f| \,\mathrm{d}\alpha.$$

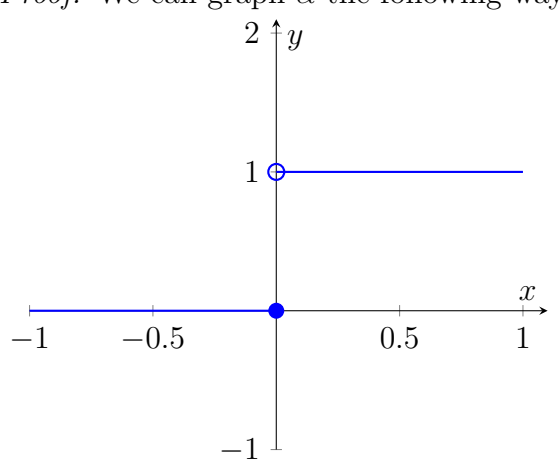
□

Theorem 6.14. Let $f : [a,b] \to \mathbb{R}$ be bounded, $s \in (a,b)$ and suppose f is continuous at s. Let

$$\alpha = \begin{cases} 0, & x \leq s \\ 1, & x > s \end{cases}$$

Then $f \in \mathcal{R}_\alpha[a,b]$ and $\int_a^b f \mathrm{d}\alpha = f(s)$.

Proof. We can graph α the following way:



[2]This is actually a good way to construct multiplication on a general ring using only addition and squaring.

Let $p = \{x_0, x_1, x_2, x_3\}$ with $x_0 = a, x_1 = s, x_3 = b$. Then

$$U(P, f, \alpha) = \sum_{n=1}^{3} M_i \Delta\alpha = M_2 = \sup_{x \in [x_1, x_2]} f(x)$$

and

$$L(P, f, \alpha) = \sum_{n=1}^{3} m_i \Delta\alpha = m_2 = \inf_{x \in [x_1, x_2]} f(x).$$

Since f is continuous at s, for all $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - y| < \delta$ then $|f(x) - f(y)| < \varepsilon/2$. So select $x_2 \in (x_1, x_1 + \delta)$. Then

$$M_2 = \sup_{x \in [x_1, x_2]} f(x) \leq f(x_1) = \varepsilon/2$$

and

$$m_2 = \inf_{x \in [x_1, x_2]} f(x) \leq f(x_1) - \varepsilon/2.$$

So $M_2 - m_2 \leq \varepsilon$. □

Remark. This is also true for

$$\alpha = \begin{cases} 0, & x < s \\ 1, & x \geq s \end{cases}$$

and for

$$\alpha = \begin{cases} 0, & x < s \\ 0.5, & x = s \\ 1, & x > s \end{cases}$$

Furthermore, all their integrals agree.

Definition 6.15. We define the Heavyside step function by $I(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$.

Theorem 6.16. Let $\{c_n\}$ be a sequence of positive real numbers with finite sum. Let $[a, b]$ be an interval and let $\{s_n\}_{n=1}^{\infty} \subset (a, b)$ and

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n).$$

Let $f : [a, b] \to \mathbb{R}$ be continuous. Then

$$\int_a^b f \mathrm{d}\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

Proof. Let $R_N = \int_a^b f \mathrm{d}\alpha - \sum_{n=1}^{N} c_n f(s_n)$. Fix N and let

$$\alpha_1 = \sum_{n=1}^{N} c_n I(x - s_n) \text{ and } \alpha_2 = \sum_{n=N}^{\infty} c_n I(x - s_n).$$

By Theorem (6.12) e,

$$\begin{aligned}\int_a^b f \mathrm{d}\alpha &= \int_a^b f \mathrm{d}\alpha_1 + \int_a^b f \mathrm{d}\alpha_2 \\ &= \sum_{n=1}^{N} \int_a^b f(x) \mathrm{d}[c_n I(x - s_n)] + \int_a^b f \mathrm{d}\alpha_2 \\ &= \sum_{n=1}^{N} c_n f(s_n) + \int_a^b f \mathrm{d}\alpha_2.\end{aligned}$$

So $R_N = \int_a^b f \mathrm{d}\alpha_2$.
Let $k = \sup_{x \in [a,b]} \{|f|\} x$, by Theorem (6.12) b,

$$\int_a^b f \mathrm{d}\alpha_2 \leq k \int_a^b \mathrm{d}\alpha_2 = k(\alpha_2(b) - \alpha_2(a)) = k \sum_{n=N+1}^{\infty} c_n.$$

It is easy to show this goes to 0 as N goes to ∞. □

The following property is closely related to the change of variables formula but is not quite the same.

Theorem 6.17. Let $f : [a, b] \to \mathbb{R}$ be bounded and $\alpha : [a, b] \to \mathbb{R}$ differentiable and monotone increasing.
Suppose $f \in \mathcal{R}_\alpha[a, b]$.
Then

$$f \in \mathcal{R}_\alpha[a, b] \Leftrightarrow f\alpha' \in \mathcal{R}[a, b]$$

and if so,

$$\int_a^b f \mathrm{d}\alpha = \int_a^b f\alpha' \mathrm{d}x.$$

Proof. Step 1: It suffices to show that

$$\overline{\int_a^b} f \mathrm{d}\alpha = \overline{\int_a^b} f\alpha' \mathrm{d}x$$

and

$$\underline{\int_a^b} f \mathrm{d}\alpha = \underline{\int_a^b} f\alpha' \mathrm{d}x.$$

We will prove the first equality and the second will be similar.
Step 2: Since $\alpha' \in \mathcal{R}[a.b]$, for all $\varepsilon > 0$, there exists P such that

$$U(P, \alpha') - L(P, \alpha') < \varepsilon.$$

This will also hold for any refined P' of P.
Then

$$\sum_{i=1}^{n} (A_i - a_i)\Delta x_i$$

cj where $A_i = \sup\{\alpha'(x) : x \in [x_{i-1}, x_i]\}$ and $a_i = \inf\{\alpha'(x) : x \in [x_{i-1}, x_i]\}$.
By the mean value theorem, for each $i \in [1, n]$, there is $t_i \in [x_{i-1}, x_i]$ such that $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$.
For every $s_i \in [x_{i-1}, x_i]$, we know

$$|\alpha'(s_i) - \alpha'(t_i)| \leq A_i - a_i$$

so,

$$\sum_{i=1}^{n} |\alpha'(s_i) - \alpha'(t_i)| \, \Delta x_i \leq sum_{i=1}^{n} |A_i - a_i| \, \Delta x_i < \varepsilon$$

for every choice of $s_i \in [x_{i-1}, x_i], i = 1, \ldots, n$.
Let $K = \sup\{|f(x)| : x \in [a, b]\}$ then

$$\sum_{i=1}^{n} |f(s_i)\Delta\alpha_i - f(s_i)\alpha'(s_i)\Delta x_i| \leq K\varepsilon.$$

Hence

$$\begin{aligned}\sum_{i=1}^{n} f(s_i)\Delta\alpha_0 &\leq \sum_{i=1}^{n} f(s_i)\alpha'(s_i)\Delta x_i + K\varepsilon \\ &\leq U(P, f\alpha') + K\varepsilon\end{aligned}$$

Taking the supremum of $s_i \in [x_{i-1}, x_i]$, $i = 1, \ldots, n$, then

$$\sum_{i=1}^{n} M_i \Delta\alpha_i \leq U(P, f\alpha') + K\varepsilon$$

thus

$$\overline{\int_a^b} f \mathrm{d}\alpha \leq U(P, f, \alpha) \leq U(P, f\alpha') + K\varepsilon.$$

Taking the infimum over all refinements of P,

$$\overline{\int_a^b} f \mathrm{d}\alpha \leq \inf_{P'} U(P, f\alpha') + K\varepsilon$$

i.e. for all $\varepsilon >$, $\overline{\int_a^b} f \mathrm{d}\alpha \leq \overline{\int_a^b} f\alpha' \mathrm{d}x + K\varepsilon$ so $\overline{\int_a^b} f \mathrm{d}\alpha \leq \overline{\int_a^b} f\alpha' \mathrm{d}x$.
The reverse inequality holds with a similar argument. $\square$

Theorem 6.19. Let $\phi : [A, B] \to [a, b]$ be strictly increasing, onto and continuous.
Let $\alpha : [a, b] \to \mathbb{R}$ be monotone increasing and $f \in \mathcal{R}_\alpha[a, b]$.
Defining $g = f \circ \phi : [A, B] \to \mathbb{R}$ and $\beta = \alpha \circ \phi : [A, B] \to \mathbb{R}$. Then $g \in \mathcal{R}_\alpha[A, B]$ and

$$\int_A^B g \mathrm{d}\beta = \int_a^b f \mathrm{d}\alpha.$$

Proof. Partitions P of $[a, b]$ and Q of $[A, B$ are in one-to-one correspondence.
We have $\alpha(x_i) = \alpha \circ \phi(y_i) = \beta(y_i)$ and

$$\{f(x)|x \in [x_{i-1}, x_i]\} = \{g(x)|y \in [y_{i-1}, y_i]\}$$

which implies

$$U(P, f, \alpha) = U(Q, g, \beta) \text{ and } L(P, f, \alpha) = L(Q, g, \beta).$$

Let $\varepsilon > 0$ then there exists a partition P of $[a, b]$ such that $U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$ since $f \in \mathcal{R}_\alpha[a, b]$. Hence by the equation above,

$$U(Q, g, \beta) - L(Q, g, \beta) < \varepsilon.$$

Thus $g \in \mathcal{R}_\beta[A, B]$.
Finally,

$$\begin{aligned}\int_A^B g \mathrm{d}\beta &= \inf_Q U(Q, g, \beta \\ &= \inf_P U(P, f, \alpha \\ &= \int_a^b f \mathrm{d}\alpha\end{aligned}$$

□

Theorem 6.20. Let $f \in \mathcal{R}_\alpha[a, b]$. For $x \in [a, b]$ define

$$F(x) = \int_a^x f(t)\mathrm{d}t \text{ and } F(a) = 0.$$

Then F is continuous on $[a, b]$. And if $c \in [a, b]$ and f is continuous at c, then F is differentiable at c with $F'(c) = f(c)$.

Proof. Let $K = \sup\{|f(t)| : t \in [a, b]\}$. By Theorem (6.12 c), for $a \leq x \leq y \leq b$,

$$F(y) - F(x) = \int_a^y f(t)\mathrm{d}t - \int_a^x f(t)\mathrm{d}t = \int_x^y f(t)\mathrm{d}t.$$

Thus

$$\begin{aligned}|F(y)-F(x)| &= \left|\int_x^f \mathrm{d}t\right| \\ &\leq \int_x^y |f|\,\mathrm{d}t \\ &\leq \int_x^y K\mathrm{d}t \\ &= K(y-x)\end{aligned}$$

Hence for every $\varepsilon > 0$, let $\delta = \dfrac{\varepsilon}{K}$ (or $\mathrm{d} = \varepsilon$ if $K = 0$) then if $|x-y| < \delta, |F(x)-F(y)| < \varepsilon$.
Now suppose $c \neq b$, if f is continuous at c, then for $h > 0$ we have

$$\begin{aligned}\left|\frac{1}{h}[F(c+h)-F(c)]-f(c)\right| &= \left|\frac{1}{h}\int_c^{c+j} f\mathrm{d}t - f(c)\right| \\ &= \frac{1}{h}\left|\int_c^{c+j} f\mathrm{d}t - \int_c^{c+h} f(c)\mathrm{d}t\right| \\ &= \frac{1}{h}\left|\int_c^{c+h} f(t)-f(c)\mathrm{d}t\right| \\ &= \frac{1}{h}\int_c^{c+h} |f(t)-f(c)|\,\mathrm{d}t\end{aligned}$$

Since f is continuous at c, for all $\varepsilon > 0$ there exists $\delta > 0$ such that for all $y \in [a,b]$ with $|y-c| < \delta$, we have $|f(c)-f(y)| < \delta$, so for $h < \delta$ we have

$$\frac{1}{h}\int_c^{c+h} |f(t)-f(c)|\,\mathrm{d} < \frac{1}{h}\int_c^{c+h} \varepsilon\ \mathrm{d}t = \varepsilon$$

So $\lim\limits_{h\searrow 0} \dfrac{F(c+h)-F(c)}{h} = f(c)$.
If $c \neq a$, a identical argument shows $\lim\limits_{h\nearrow 0} \dfrac{F(c+h)-F(c)}{h} = f(c)$. □

Remark. If f is not continuous at c, is it the case that F can never be differentiable at c ?
The answer no. If $f(x) = g(x)$ except at c, then $\displaystyle\int_a^b f\mathrm{d}x = \int_a^b g\mathrm{d}x$ and $F'(c) = \lim\limits_{x\to c} f(x)$.
However if f has a jump discontinuity at c, then F is not differentiable.

Theorem 6.21 (Fundamental Theorem of Calculus). If $f \in \mathcal{R}_\alpha[a,b]$ and there is a differentiable function F on $[a,b]$ such that $F' = f$, then

$$\int_a^b f\mathrm{d}x = F(b)-F(a).$$

Proof. By the MVT, take any partition of $P = \{a = x_0, \dots, x_n = b\}$ then there are $t_1, \dots, t_n$ where $t_i \in [x_{i-1}, x_i]$, $i = 1, \dots, n$ such that $\Delta x_i F'(t_i) = F(x_i) - F(x_{i-1})$ so

$$\begin{aligned} F(b) - F(a) &= \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) \\ &= \sum_{i=1}^{n} F'(t_i)\Delta x_i \in [L(P, F'), U(P, F')]. \end{aligned}$$

Note that $[L(P, F'), U(P, F')] = [L(P, f), U(P, f)]$ and $\int_a^b f \mathrm{d}x \in [L(P, f), U(P, f)]$.
Hence

$$\left| \int_a^b f \mathrm{d}x - (F(b) - F(a) \right| \leq U(P, f) - L(P, f).$$

Since $f \in \mathcal{R}[a, b]$, for all $\varepsilon > 0$ there exists a partition P such that $U(P, f) - L(P, f) < \varepsilon$, so

$$\left| \int_a^b f \mathrm{d}x - (F(b) - F(a)) \right| \leq \varepsilon.$$

□

Theorem 6.22 (Integration by Parts). Let $F, G : [a, b] \to \mathbb{R}$ be differentiable. Let $f = F'$ and $g = G'$ and suppose $f, g \in \mathcal{R}[a, b]$. Then

$$\int_a^b F(x)g(x)\mathrm{d}x = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)\mathrm{d}x.$$

Proof. Let $H(x) = F(x)G(x)$. Then $H'(x) = f(x)G(x) + F(x)g(x) \in \mathcal{R}[a, b]$ by Theorems (6.13) and (6.12).
Then by Theorem (6.21),

$$H(b) - H(a) = \int_a^b H'(x)\mathrm{d}x$$

i.e.

$$F(b)G(b) - F(a)G(a) = \int_a^b f(x)G(x)\mathrm{d}x + \int_a^b F(x)g(x)\mathrm{d}x$$

□

If $F : [a, b] \to \mathbb{R}$ is differentiable and $F' = f$, do we need $f \in \mathcal{R}[a, b]$ or does this hold automatically ?
Well... a more complicated question... is there $F : [a, b] \to \mathbb{R}$ differentiable such that F' is discontinuous at every point in $[a, b]$? The answer to this question is no.
Then what sets of $[0, 1]$ can be the set of discontinuities of f' where f is differentiable ? These are all the sets of measure 0. We will see this a bit later in the course.
Back to the original question, if $f : [a, b] \to \mathbb{R}$ is differentiable with f' unbounded then this might not hold.

Take $F(x) = \begin{cases} 0 \text{ if } x = 0 \\ x^2 \sin(\frac{1}{x}) \text{ if } x \neq 0 \end{cases}$. Then f is differentiable. However on the sequence $x_n = \frac{1}{\sqrt{\pi n}}$, $F'(x_n)$ blows up. So $F' \notin \mathcal{R}[a,b]$.
Another way for this would be if F' is bounded but F' is discontinuous on a set of positive measure.

Definition 6.23. If $f : [a, \infty) \to \mathbb{R}$ satisfies $f \in \mathcal{R}[a,b]$ for all $b > a$ then we define

$$\int_a^\infty f \mathrm{d}x = \lim_{b\to\infty} \int_a^b f \mathrm{d}x.$$

If this limit exists, we say the integral $\int_a^\infty f \mathrm{d}x$ converges.
If $\int_a^\infty |f| \, \mathrm{d}x$ exists, we say $\int_a^\infty f \mathrm{d}x$ converges absolutely.
The same idea goes for

$$\int_{-\infty}^b f \mathrm{d}x$$

.

Definition 6.24. If $f : \mathbb{R} \to \mathbb{R}$ and both

$$\int_0^\infty f \mathrm{d}x \text{ and } \int_{-\infty}^0 f \mathrm{d}x$$

converges (absolutely) then we define

$$\int_{-\infty}^\infty f \mathrm{d}x = \int_0^\infty f \mathrm{d}x + \int_{-\infty}^0 f \mathrm{d}x$$

and we say $\int_{-\infty}^\infty f \mathrm{d}x$ converges (absolutely).

Is it possible to construct a function where $\int_{-\infty}^\infty f \mathrm{d}x$ converges but not absolutely ?

Yes. Take $f(x) = \frac{1}{\lceil x \rceil (-1)^{\lceil x \rceil}}$.

6.3 Integration of complex valued functions

Definition 6.25. Let $f : [a,b] \to \mathbb{C}$ and $\alpha : [a,b] \to \mathbb{R}$ be monotone increasing.
We say $f \in \mathcal{R}_\alpha[a,b]$ if $\mathrm{Re}(f) \in \mathcal{R}_\alpha[a,b]$ and $\mathrm{Im}(f) \in \mathcal{R}_\alpha[a,b]$. And if so, we define

$$\int_a^b f \mathrm{d}\alpha := \int_a^b \mathrm{Re}(f) \mathrm{d}\alpha + i \int_a^b \mathrm{Im}(f) \mathrm{d}\alpha$$

Same idea if f takes values in a finite dimension, real or complex valued vector space.

Chapter 7

Sequences and Series of Functions

7.1 Discussion of main problem

Let E be a set and $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions $f_n : E \to S$ (S is usually $\mathbb{R}$ or $\mathbb{C}$).
Let $f : E \to S$.
What does it mean for the sequence f_n to converge to f ?
If $f_n \to f$ and the f_n all have some property, must it be true that the limiting function f also has the that property ?
We will see it depends on what type of convergence we are using.

Definition 7.1. Let E be a set and (M, d) be a metric space. Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions $f_n : E \to M$. If the sequence $\{f_n\}_{n=1}^{\infty}$ converges for every $x \in E$, then we say that $\{f_n\}$ converges pointwise on E.
If this happens, we can define

$$f(x) = \lim_{n\to\infty} f_n(x),\ f : E \to M$$

and we say $f_n \to f$ pointwise (on E).
i.e. $\forall x \in E, \forall \varepsilon > 0, \exists N$ s.t. $\forall n > N, d(f_n(x) - f(x)) < \varepsilon$

Example. Let $E = \mathbb{R}$, $M = \mathbb{R}$ with the usual metric.

i) If each f_n is continuous and $f_n \to f$ p.w. must ie be true that f is continuous ?

   Take $f_n = e^{-nx}$. Then $f_n \to f = \begin{cases} 1, \text{ if } x = 0 \\ 0, \text{ if } x \neq 0 \end{cases}$

   Since f is not continuous we see the claim to be false.

ii) If each f_n is bounded and $f_n \to f$ pointwise, must f be bounded ?

   Take $f_n = \begin{cases} -n, & \text{if } x \in (-\infty, n] \\ x, & \text{if } x \in [-n, n] \\ n, & \text{if } x \in [n, \infty) \end{cases}$

   Then each f_n is bounded but $f_n \to f = x$ pointwise which is not bounded. So the claim is false.

iii) If each f_n is bounded by 1, i.e. $|f_n(x)| \leq 1$ for all $x \in E$, and $f_n \to f$ pointwise, must $|f(x)| \leq 1$ for all $x \in E$?

Yes.

If $f(x) > 1$ for some x, $f(x)-1 > 0$ so there is some n such that $|f_n(x)-f(x)| < f(x)-1$ which implies $f_n(x) > 1$ which is a contradiction.

As we can see some properties will remain for the limiting function and some properties do not remain. This tells us we might need a stronger condition that pointwise convergence if we want these properties to remain.
In general we can see that many of the properties that do not hold are because of one problem :

$$\lim_{n\to\infty}\lim_{m\to\infty} a_{m,n} \text{ is not always equal to } \lim_{m\to\infty}\lim_{n\to\infty} a_{m,n}$$

Example. Let $a_{m,n} = \dfrac{n}{n+m}$. Then

$$\lim_{n\to\infty} a_{n,m} = 1 \Rightarrow \lim_{m\to\infty}\lim_{n\to\infty} a_{m,n} = 1$$

and

$$\lim_{m\to\infty} a_{n,m} = 0 \Rightarrow \lim_{n\to\infty}\lim_{m\to\infty} a_{m,n} = 0.$$

7.2 Uniform convergence

Definition 7.7. Let E be a set and (M, d) a metric space. Let f_n be a sequence of functions $f_n : E \to M$ and let $f : E \to M$ be a function. We say $f_n \to f$ uniformly if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ if for all $n > N, x \in E$,

$$d(f_n(x) - f(x)) < \varepsilon.$$

Uniform convergence is nice because it conserves most of the "nice" properties we like, that aren't conserved by pointwise convergence.

Theorem 7.8 (Cauchy criteria for uniform convergence). Let E be a set and (M, d) a complete metric space. Let $\{f_n\}$ be a sequence $f_n : E \to M$.
Then $\{f_n\}$ converges uniformly to f if and only if for all ε there exists $N \in \mathbb{N}$ such that for all $n, m > N, x \in E$,

$$d(f_n(x), f_m(x)) < \varepsilon. \tag{*}$$

Proof. ($\Rightarrow$) Suppose $\{f_n\} \to f$ uniformly. Let $\varepsilon > 0$ then there exists N such that for all $n > N, x \in E$,

$$d(f_n(x), f(x)) \leq \frac{\varepsilon}{2}.$$

Then for all $m, n > N$,

$$d(f_n(x), f_m(x)) \leq d(f_n(x), f(x)) + d(f(x), f_m(x)) < \varepsilon.$$

($\Leftarrow$) Suppose that $\{f_n\}$ satisfies (*). Then for each $x \in E$, $f_n(x)$ is a Cauchy sequence in M and (M, d) is complete so $f_n(x)$ converges.
Define $f(x) = \lim_{n\to\infty} f_n(x)$ and let $\varepsilon > 0$. Since f_n satisfies (*), we know there is N such that for all $n, m > N$ and $x \in E$,

$$d(f_n(x), f_m(x)) < \frac{\varepsilon}{2}.$$

Hence

$$d(f_n(x), f(x)) \leq d(f_n(x), f_m(x)) + d(f_m(x), f(x))$$

Since $f_m(x) \to f(x)$ we can select $m > N$ s.t

$$d(f_m(x), f(x)) < \frac{\varepsilon}{2}.$$

Hence

$$d(f_n(x), f(x)) < \varepsilon.$$

□

Idea of Proof. $\varepsilon/2$

Theorem 7.9. Suppose $\lim_{n\to\infty} f_n(x) = f(x)$ and put

$$M_n = \sup_{x\in E} d(f_n(x) - f(x)).$$

Then $f_n \to f$ uniformly if and only if $M_n \to 0$ as $n \to \infty$.

Proof. This proof is immediate from the definition of uniform convergence. □

Theorem 7.10 (Weierstrass M-test). Suppose $\{f_n\}$ is a sequence of functions on E, and suppose

$$|f_n(x)| \leq M_n \qquad (x \in E, n = 1, 2, 3, \dots).$$

Then $\sum f_n$ converges uniformly[1] if $\sum M_n$ converges.

Proof. If $\sum M_n$ converges, then, for abritrary $\varepsilon > 0$,

$$\left|\sum_{i=n}^{m} f_i(x)\right| \leq \sum_{i=n}^{m} M_i \leq \varepsilon \qquad (x \in E),$$

provided m and n are large enough. Hence by Theorem 7.8, $f_n \to f$ uniformly. □

[1]We say $\sum f_n$ converges pointwise (resp. uniformly) if the sequence $s_n = \sum_{i=1}^{n} f_i$ converges pointwise (resp. uniformly).

7.3 Uniform convergence and continuity

Theorem 7.11. Let (M_1, d_1) and (M_2, d_2) be metric spaces with (M_2, d_2) complete. Let $E \subseteq M_1$, let $\{f_n\}$ be a sequence of functions $f_n : E \to M_2$ and suppose $\{f_n\} \to f$ uniformly on E. Let x be a limit point of E and suppose $\lim_{t\to x} f_n(t) = y_n$ exists for each n.
Then $\{y_n\}$ is a convergent sequence i.e. $y_n \to y$ and $\lim_{t\to x} f(t) = y$.
Another formulation of the conclusion is

$$\lim_{t\to x}\lim_{n\to\infty} f_n(t) = \lim_{n\to\infty}\lim_{t\to x} f_n(t).$$

Proof. Step 1: Show that $y_n \to y$ for some $y \in M_2$. Since M_2 is complete, it suffices to show that y_n is Cauchy.
Let $\varepsilon > 0$. Choose N large enough such that for all $x \in E$,

$$d_2(f_n(x), f_m(x)) < \frac{\varepsilon}{3}.$$

Then,

$$d_2(y_n, y_m) \leq d_2(y_n, f_n(t)) + d_2(f_n(t), f_m(t)) + d_2(f_m(t), y_m)$$

Then since then we can choose t close enough to x such that

$$d_2(f_n(t), y_n) < \frac{\varepsilon}{3} \text{ and } d_2(f_m(t), y_m) < \frac{\varepsilon}{3}.$$

Hence

$$d_2(y_n, y_m) < \varepsilon.$$

Step 2: Show that $\lim_{t\to x} f(t) = y$. For all $t \in E$ and n we have,

$$d_2(f(t), y) \leq d_2(f(t), f_n(t)) + d_2(f_n(t), y_n) + d_2(y_n, y).$$

Let $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists N_1 such that for all $n > N_1$ and $t \in E$,

$$d_2(f(t), f_n(t)) < \frac{\varepsilon}{3}.$$

Similarly, since $y_n \to y$, there exists N_2 such that for all $n > N_2$,

$$d_2(y_n, y) < \frac{\varepsilon}{3}.$$

Let $N = \max(N_1, N_2)$. Then for all $n > N$,

$$d_2(f(t), f_n(t)) + d_2(y_n, y) < \frac{2\varepsilon}{3}.$$

Then for each n, there exists δ such that if $d_1(t, x) < \delta$, then

$$d_2(f_n(t), y_n) < \frac{\varepsilon}{3}.$$

Hence for all $t \in E$ with $d_1(t, x) < \delta$ and $n > N$,

$$d_2(f(t), y) < \varepsilon.$$

□

Idea of Proof. $\varepsilon/3$

Corollary 7.12. Let $(M_1, d_1), (M_2, d_2), \{f_n\}, f$ and E be as before. If each $\{f_n\}$ is continuous on E and $\{f_n\} \to f$ uniformly, then f is continuous on E.

Proof. f is always continuous at isolated points, so it suffices to prove f is continuous at limit points $x \in E \cap E'$.[2]
For every such x, theorem 7.11 implies

$$f(x) = \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \lim t \to x f_n(t) = \lim t \to x f(t)$$

□

Theorem 7.13 (Dini's uniform convergence Theorem). Let (M, d) be a compact metric space, $\{f_n\}$ a sequence of functions $f_n : M \to \mathbb{R}$. Suppose

- Each f_n is continuous on M.
- f_n converges pointwise to some $f : M \to \mathbb{R}$ which is continuous.
- $f_{n+1}(x) \geq f_n(x)$ for each $x \in M$, $n \in \mathbb{N}$.

Then $f_n \to f$ uniformly on M.

Proof. Let $g_n = f - f_n$. Then

- g_n is continuous.
- $g_n \to 0$ pointwise.
- $g_n \geq g_{n+1} > 0$ for all n.

We want to show $g_n \to 0$ uniformly. Since g_n is decreasing, it suffices to show for every ε there exists N such that for all $n > N, x \in M$,

$$|g_n(x)| < \varepsilon.$$

Let K_n be the set of all $x \in M$ such that $g(x) \geq \varepsilon$. Since g_n is continuous, K_n is closed[3] and is it is compact since M is compact[4]. Since g_n is decreasing,

$$K_{n+1} \subseteq K_n.$$

Since $g_n \to$ pointwise, for each $x \in M$, there is n such that

$$g_n(x) < \varepsilon.$$

[2] E' is the closure of E
[3] The pre-image of closed interval under a continuous function is continuous.
[4] A closed subset of a compact metric space is compact.

That is $x \notin K_n$. Since x was arbitrary,

$$\bigcap_{n=1}^{\infty} K_n = \emptyset.$$

Hence by Theorem 2.36, there exists N such that $K_N = \emptyset$.
That is,

$$g_N(x) < \varepsilon$$

for all $x \in M$. Therefore

$$|g_n(x)| < \varepsilon$$

for all $x \in M$ and $n \geq N$. □

Idea of Proof. Find set of points where distance of $f_n(x)$ to $f(x)$ is large. Then this must be a closed set, nested and eventually empty.

Definition 7.14. Let (X, d) be a metric space. Define

$$\mathcal{C}(X) = \{f : X \to \mathbb{C} \mid f \text{ is bounded and continuous.}\}$$

(Remark: If X is compact, every f is bounded anyways).
For each $f \in \mathcal{C}(X)$, we can define the "supremum norm"

$$||f|| = \sup_{x \in X} |f(x)|.$$

For all $f \in \mathcal{C}(X)$, this is finite. There exist alternate notations

$$||f||_{\mathcal{C}(X)}, \quad ||f||_{\mathcal{C}'(X)}, \quad ||f||_{\infty}$$

Then $\mathcal{C}(X)$ is a complex vector space with norm $||\cdot||$.
Indeed, for all $f, g \in \mathcal{C}(X), \lambda \in \mathbb{C}$,

- $||f|| \geq 0$ with $||f|| = 0 \Leftrightarrow f(x) = 0 \ \forall x \in X$.
- $||\lambda f|| = |\lambda| \, ||f||$.
- $|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq ||f|| + ||g|| \Rightarrow ||f + g|| \leq ||f|| + ||g||$.

Thus if $\rho(f, g) = ||f - g||$ then ρ is a metric and $(\mathcal{C}(X), \rho)$ is a metric space.
Then

$$\begin{aligned} f_n \to f \text{ uniformly} &\Leftrightarrow \forall \varepsilon > 0, \exists N \text{ s.t. } \forall x \in X, n \geq N, |f_n(x) - f(x)| < \varepsilon \\ &\Leftrightarrow \forall \varepsilon > 0, \exists N \text{ s.t. } n \geq N, ||f_n - f|| < \varepsilon \\ &\Leftrightarrow f_n \to f \text{ in } \mathcal{C}(X). \end{aligned}$$

Theorem 7.15. $\mathcal{C}(X)$ is complete.

Proof. Let $\{f_n\}$ be a Cauchy sequence in $\mathcal{C}(X)$. By the Cauchy criteria, $f_n \to f$ uniformly for some $f : X \to \mathbb{C}$. Then by Corollary 7.12, f is continuous.
Since $f_n \to f$ uniformly, there exists N such that

$$|f_N(x) - f(x)| < 1$$

for all $x \in X$, so

$$|f(x)| < |f_N(x)| + 1 \leq ||f_N|| + 1$$

This implies that

$$||f(x)|| \leq ||f_N(x)|| + 1 < \infty$$

So f is bounded, hence $f \in \mathcal{C}(X)$. □

Idea of Proof. $f(x)$ must be relatively close to $f_n(x)$ which is bounded.

What about if we're working with $\mathcal{C}(X, Y)$ which is the space of all bounded continuous functions $f : X \to Y$?
Then f being bounded means $f(x) \in B_r(y)$ for some finite $r, y \in Y$.
Let $d(f, g) = \sup\limits_{x \in X} d_Y(f(x), g(x))$.
Then $\mathcal{C}(X, Y)$ is complete if and only if Y is complete. In fact the previous proof works by replacing $|x, y|$ with $d_Y(x, y)$.

7.4 Uniform convergence and integration

Theorem 7.16. Let $\alpha : [a, b] \to \mathbb{R}$ be monotone increasing. Let $\{f_n\}$ be a sequence, $f_n \in \mathcal{R}_\alpha[a, b]$ for $n = 1, 2, 3, \ldots$. Let $f : [a, b] \to \mathbb{R}$ and $f_n \to f$ uniformly.
Then $f \in \mathcal{R}_\alpha[a, b]$ and

$$\lim_{n \to \infty} \int_a^b f_n \mathrm{d}\alpha = \int_a^b f \mathrm{d}\alpha.$$

Proof. Step 1: Show $f \in \mathcal{R}_\alpha[a, b]$.
Let $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists N such that for all $n > N$, $x \in [a, b]$,

$$|f(x) - f_n(x)| < \varepsilon.$$

So,

$$f_n(x) - \varepsilon \leq f(x) \leq f_n(x) + \varepsilon.$$

Thus,

$$\int_a^b f_n(x) - \varepsilon \, \mathrm{d}\alpha \leq \underline{\int_a^b} f(x) \mathrm{d}\alpha \leq \overline{\int_a^b} f(x) \mathrm{d}\alpha \leq \int_a^b f_n(x) + \varepsilon \, \mathrm{d}\alpha. \tag{1}$$

We can rearrange this as

$$\begin{aligned} \overline{\int_a^b} f(x) \mathrm{d}\alpha - \underline{\int_a^b} f(x) \mathrm{d}\alpha &\leq \int_a^b f_n(x) + \varepsilon \, \mathrm{d}\alpha - \int_a^b f_n(x) - \varepsilon \, \mathrm{d}\alpha \\ &= \int_a^b 2\varepsilon \, \mathrm{d}\alpha = 2\varepsilon(\alpha(b) - \alpha(a)). \end{aligned}$$

Since ε was arbitrary, we have

$$\overline{\int_a^b} f(x)\mathrm{d}\alpha = \underline{\int_a^b} f(x)\mathrm{d}\alpha.$$

Step 2: Show $\displaystyle\int f_n \mathrm{d}\alpha \to \int f \mathrm{d}\alpha$.
Rearranging (1) gives

$$\int_a^b f_n \mathrm{d}\alpha - \int_a^b \varepsilon \,\mathrm{d}\alpha \leq \int_a^b f \mathrm{d}\alpha \leq \int_a^b f_n \mathrm{d}\alpha + \int_a^b \varepsilon \mathrm{d}\alpha.$$

Which implies

$$\left| \int_a^b f_n \mathrm{d}\alpha - \int_a^b f \mathrm{d}\alpha \right| \leq \varepsilon(\alpha(b) - \alpha(a)).$$

□

Idea of Proof. Find f_n everywhere within ε of f.

Note. Uniform convergence is needed because we might not find such a function. Here is a counter-example for why this doesn't work for pointwise convergence.
Let $a = -1, b = 1$ and set $f_n : [-1, 1] \to \mathbb{R}$ as

$$f_n = \begin{cases} 2n, & \text{if } x \in \left[-\dfrac{1}{n}, 0\right) \cup \left(0, \dfrac{1}{n}\right] \\ 0, & \text{otherwise} \end{cases}$$

Clearly, $f_n(x) \to 0$ for all x by the archimedean property.
However, for all n, $\displaystyle\int_{-1}^1 f_n(x)\mathrm{d}x = 1 \not\to \int_{-1}^1 0 \,\mathrm{d}x = 0$.

Corollary 7.17. If $f_n \in \mathcal{R}_\alpha[a, b]$ and $\displaystyle\sum_{n=1}^\infty f_n$ converges uniformly on $[a, b]$ to f, then $f \in \mathcal{R}_\alpha[a, b]$ and

$$\int_a^b f \mathrm{d}\alpha = \sum_{n=1}^\infty \int_a^b f_n \mathrm{d}\alpha.$$

Proof. Let $g_n = \displaystyle\sum_{i=1}^n f_i$, then $g_n \to f$ uniformly and we can apply Theorem 7.16. □

7.5 Uniform convergence and differentiation

Theorem 7.18. Let f_n be a sequence $f_n : [a, b] \to \mathbb{R}$. Suppose that :

- Each f_n is differentiable on $[a, b]$.

- There exists $x_0 \in [a,b]$ such that $\{f_n(x_0)\}$ converges.
- f_n' converges uniformly on $[a,b]$.

Then $\{f_n\}$ converges uniformly on $[a,b]$ to a function f, and

$$f'(x) = \lim_{n\to\infty} f_n'(x) \qquad (x \in [a,b]).$$

Proof. Step 1: $f_n \to f$ uniformly.
Let $\varepsilon > 0$ and let N large enough that

$$|f_n(x_0) - f_m(x_0)| < \varepsilon \qquad (m,n \geq N)$$

and

$$|f_n'(x) - f_m'(x)| < \varepsilon \qquad (m,n > N, x \in [a,b]).$$

We can then apply the MVT to $(f_n - f_m)$: for $x,t \in [a,b], x \neq t$, there exists c between x and t such that

$$|(f_n(x) - f_m(x)) - (f_n(t) - f_m(t))| = |(f_n' - f_m')(c)||x-t| < \varepsilon|x-t|.$$

This has two consequences:

$$|(f_n(x) - f_m(x)) - (f_n(t) - f_m(t))| \leq \varepsilon|b-a| \tag{1}$$

$$\frac{|f_n(x) - f_m(x) - f_n(t) + f_m(t)|}{|x-t|} < \varepsilon. \tag{2}$$

By (1), for $n,m \geq N, x \in [a,b], t = x_0$

$$|f_n(x) - f_m(x)| \leq |(f_n(x) - f_m(x)) - (f_n(x_0) - f_m(x_0) + (f_n(x_0) - f_m(x_0))| < \varepsilon(b-a) + \varepsilon$$

Hence $\{f_n\}$ satisfies the Cauchy criteria for uniform convergence. We can then define $f = \lim\limits_{n\to\infty} f_n$.
Step 2: $f' = \lim\limits_{n\to\infty} f_n'$.
Fix $x \in [a,b]$. Define $\phi_n(t) = \dfrac{f_n(t) - f_n(x)}{t-x}$ and $\phi(t) = \dfrac{f(t) - f(x)}{t-x}$.
So, $f_n'(x) = \lim\limits_{t\to x} \phi_n(t)$ and the same goes for $f'(x)$ if it exists. From (2) we have for all $n,m \geq N$,

$$|\phi_n(t) - \phi_m(t)| < \varepsilon.$$

Hence $\{\phi_n\}$ satisfy the Cauchy criteria for uniform convergence. Hence ϕ_n converges uniformly on $[a,b]\backslash\{x\}$. Fix $t \in [a,b]\backslash\{x\}$. Then

$$\phi_n(t) - \phi(t) = \left|\frac{f_n(t) - f_n(x) - f(t) + f(x)}{t-x}\right| \leq \left|\frac{f_n(t) - f(t)}{t-x}\right| + \left|\frac{f_n(x) - f(x)}{t-x}\right|$$

which goes to 0 as $n \to \infty$.
So $\phi_n \to \phi$ pointwise and hence uniformly on $[a,b]\backslash\{x\}$. We then use Theorem 7.11 to conclude

$$f'(x) = \lim_{t\to x} \phi(t) = \lim_{n\to\infty} \lim_{t\to x} \phi_n(x) = \lim_{n\to\infty} f_n'(x).$$

□

Idea of Proof. By MVT f_n is Cauchy. Then derivative is converges uniformly.

Theorem 7.19. There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that $f'(x)$ does not exists at any point $x \in \mathbb{R}$.

Proof. Let $\phi(x)$ be the periodic extension of $g(x) : [-1, 1] \to \mathbb{R}$ where $g(x) = |x|$. Then $\phi(x)$ is continuous and in fact Lipschitz continuous with Lipschitz constant 1 i.e. $|\phi(x) - \phi(y)| \leq |x - y|$.
Let

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \phi(4^n x)$$

Then this series converges by the Weierstrass M-test, and furthermore $f(x)$ is continuous.
Fix $x \in \mathbb{R}$ and set $\delta_m = \pm\frac{1}{2}4^{-m}$ with the sign chosen such that no integer lies between $4^m x$ and $4^m(x + \delta_m)$. Then

$$\left|\frac{f(x+\delta_m) - f(x)}{\delta_m}\right| = \left|\sum_{n=0}^{m} \left(\frac{3}{4}\right)^n \frac{\phi(4^n(x+\delta_m) - \phi(4^n x)}{\delta_m}\right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1).$$

which goes to ∞ as $m \to \infty$. □

7.6 Equicontinuity

Definition 7.22. Let (X, d) be a metric space and let $E \subset X$. Let $\mathcal{F}$ be a family of functions $f : E \to \mathbb{C}$. We say $\mathcal{F}$ is equicontinuous if for all x, there exists δ such that for all $x, y \in E$ with $d(x, y) < \delta$, for all $f \in \mathcal{F}$,

$$|f(x) - f(y)| < \varepsilon.$$

Remark. (1) If $\mathcal{F}$ is equicontinuous, each $f \in \mathcal{F}$ is uniformly continuous.

(2) The converse is false. Take $X = [0, 1]$, $\mathcal{F} = \{x^n\}$.

(3) If $\mathcal{F}$ is finite and each $f \in \mathcal{F}$ is uniformly continuous then $\mathcal{F}$ is equicontinuous.

Theorem 7.23. Let (X, d) be a metric space $E \subset X$ countable. Let $f_n : E \to \mathbb{C}$ and suppose $\{f_n\}$ is pointwise bounded on E. Then there exists a subsequence that converges.

Proof. Write $E = \{x_1, x_2, \dots\}$. We know $\{f_n(x_1)\}_{n\in\mathbb{N}}$ is a bounded sequence of numbers, hence has a convergent subsequence $\{f_{1,k}\}_{k\in\mathbb{N}}$. Then construct successive subsequences

$$\begin{matrix} f_{1,1} & f_{1,2} & f_{1,3} & \cdots \\ f_{2,1} & f_{2,2} & f_{2,3} & \cdots \\ f_{3,1} & f_{3,2} & f_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{matrix}$$

Then $\{f_{i,k}\}_{k\in\mathbb{N}}$ is a subsequence of $\{f_{i-1,k}\}_{k\in\mathbb{N}}$. We take the diagonal subsequence $\{f_{i,u}\}_{i\in\mathbb{N}}$. This converges at x_j for every j because $\{f_{i,i}\}_{i\in\mathbb{N}}$ is a subsequence of $\{f_{j,k}\}_{k\in\mathbb{N}}$ except possibly for the first $j - 1$ elements. □

Idea of Proof. Cantor's diagonalization argument.

Theorem 7.24. Let K be a compact metric space, $f_n : K \to \mathbb{C}$ continuous. Suppose $\{f_n\}_{n=1}^{\infty}$ converges uniformly on K. Then $\{f_n\}_{n=1}^{\infty}$ is equicontinuous.

Proof. Let $\varepsilon > 0$. Since $\{f_n\}_{n=1}^{\infty}$ converges uniformly, there exists N such that for all $n, m \geq N$,

$$|f_n - f_m| \leq \frac{\varepsilon}{3}. \tag{1}$$

Since K is compact, each f_n is uniformly continuous so $\{f_1, \ldots, f_N\}$ is equicontinuous i.e. there exists δ such that for all $x, y \in K, n \leq N$ with $d(x, y) < \delta$,

$$|f_n(x) - f_n(y)| < \frac{\varepsilon}{3}.$$

Then if $n > N$ and $x, y \in K$ with $d(x, y) < \delta$,

$$|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \varepsilon.$$

□

Idea of Proof. $\varepsilon/3$

Theorem (Problem 7.16). Let K be a compact metric space and $\{f_n\}$ an equicontinuous family of functions $f_n : K \to \mathbb{C}$. If $\{f_n\}_{n=1}^{\infty}$ converge pointwise on K, then it converges uniformly.

Proof. Let $\varepsilon > 0$ and set $\delta > 0$ such that for all $x, y \in K$ with $d(x, y) < \delta$ and for all f_n,

$$|f_n(x) - f_n(y)| < \frac{\varepsilon}{3}.$$

Since K is compact the open cover $N_\delta(x)_{x \in K}$ has a finite sub-cover $\{N_\delta(x_1), \ldots, N_\delta(x_l)\}$. Then give $x \in K$, there exists x_j such that $d(x, x_j) < \delta$, so for $n, m \in \mathbb{N}$

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_n(x_j)| + |f_n(x_j) - f_m(x_j)| + |f_m(x_j) - f_m(x)|$$

Since $\{f_n\}$ converges pointwise, for each $j = 1, \ldots, l$ there exists N_j such that for all $m, n \geq N_j$,

$$|f_n(x_j) - f_m(x_j)| < \frac{\varepsilon}{3}.$$

Let $N = \max(N_1, N_2, \ldots, N_j)$. Then for all $m, n \geq N$, $j = 1, 2, \ldots, l$

$$|f_n(x_j) - f_m(x_j)| < \frac{\varepsilon}{3}.$$

Hence $|f_n(x) - f_m(x)| < \varepsilon$. □

Idea of Proof. $\varepsilon/3$ and each x must be within δ of a point where we know f_n converges pointwise.

Definition 7.19. Let X be a metric space, $E \subset X$, let $\mathcal{F}$ be a family of functions $f : E \to \mathbb{C}$. We say $\mathcal{F}$ is pointwise bounded if there exists $\phi : E \to \mathbb{R}$ such that

$$|f(x)| < \phi(x) \qquad (x \in E, f \in \mathcal{F}).$$

We say $\mathcal{F}$ is uniformly bounded if there exists M such that

$$|f(x)| < M \qquad (x \in E, f \in \mathcal{F}).$$

Note. Uniform boundedness can be extended to a arbitrary metric space (Y, ρ) by $f(x) \in B_M(\alpha)$ for some $\alpha \in Y, M \in \mathbb{R}$.

Theorem 7.25 (Arzelà-Ascoli Theorem). Let K be a compact metric space. Let $\{f_n\} \subset (C)(K)$ be equicontinuous and pointwise bounded. Then

(a) $\{f_n\}$ is uniformly bounded.

(b) $\{f_n\}$ has a uniformly convergent subsequence.

Proof. a) Since $\{f_n\}$ is equicontinuous, there exists $\delta > 0$ such that for all $n \in \mathbb{N}, x, y \in K$ with $d(x, y) < \delta$, we have

$$|f_n(x) - f_n(y)| < 1.$$

Since K is compact the open cover $N_\delta(x)_{x \in K}$ has a finite sub-cover $\{N_\delta(x_1), \ldots, N_\delta(x_l)\}$ and for each $i = 1, \ldots, l, \{f_n(x_i)\}_{n=1}^\infty$ is bounded by M_i. Let $M = 1 + \max(M_1, \ldots, M_l)$. For $x \in K, n \in \mathbb{N}$,

$$|f_n(x)| \leq |f_n(x_i)| + |f_n(x) - f_n(x_i)| \leq M_i + 1 \leq M.$$

b) Step 1: Let E be a countable, dense[5] subset of K. By Theorem 7.23, there exists a subsequence of functions that converge pointwise on E. Call this sequence $\{g_i\}$. We will show $\{g_i\}$ satisfies the Cauchy criteria on K.

Step 2: Let $\varepsilon > 0$. By equicontinuity, of $\{g_i\}$, there exists δ such that for all $x, y \in K$ with $d(x, y) < \delta$, for all $i \in \mathbb{N}$,

$$|g_i(x) - g_i(y)| < \frac{\varepsilon}{3}.$$

Since E is dense, $\{N_\delta(y)\}_{y \in E}$ is a cover of K, so there is a finite sub-cover $\{N_\delta(y_1), \ldots, N_\delta(y_l)\}$ $y_i \in E$. So for all $x \in K$, there exists k such that $d(x_k, y_k) < \delta$.

Step 3: For $x \in K$, $i, j \in \mathbb{N}$,

$$|g_i(x) - g_j(x)| \leq |g_i(x) - g_i(y_k)| + |g_i(y_k) - g_j(y_k)| + |g_j(y_k) - g_j(x)|.$$

If we choose $N \in \mathbb{N}$ large enough, for all $i, j \geq N$,

$$|g_i(y_k) - g_j(y_k)| < \frac{\varepsilon}{3} \qquad (k = 1, 2, \ldots, l).$$

□

Idea of Proof. a) Each open cover has a maximum, so there must be a greatest maximum. b) Convergent subsequence of a countable dense subset of K. x must be within δ of a point of the dense subset so we can bound the difference. The sequence if pointwise convergent: $\varepsilon/3$.

[5] E is dense in K if every open set in K intersects E.

7.7 The Stone-Weierstrass Theorem

Definition A. Let $f, g : \mathbb{R} \to \mathbb{R}$ or $\mathbb{R} \to \mathbb{C}$ be Riemann integrable. For $x \in \mathbb{R}$ we define

$$f * g(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\mathrm{d}t$$

if the integral exists. $f * g$ is a function whose domain is a subset of $\mathbb{R}$ (in most examples we care about, the domain is $\mathbb{R}$).

Claim. If $f * g(x)$ exists, then $g * f(x)$ exists and both are equal.

Proof. If $f * g(x)$ exists, then set $s = x - t$. Then

$$f * g(x) = \int_{-\infty}^{\infty} f(t)g(x-t)\mathrm{d}t = \int_{\infty}^{-\infty} f(x-s)g(s)\mathrm{d}(-s) = \int_{-\infty}^{\infty} g(s)f(x-s)\mathrm{d}(s) = g * f(x).$$

□

Definition B. We say that a sequence of functions $\{f_n\}$, $f_n : \mathbb{R} \to \mathbb{C}$ is called an approximate identity if they satisfy the following:

(a) $\displaystyle\int_{-\infty}^{\infty} f_n(t)\mathrm{d}t = 1$ for all $n \in \mathbb{N}$.

(b) There exists $M \geq 0$ such that $\displaystyle\int_{-\infty}^{\infty} |f_n(t)|\mathrm{d}t \leq M$ for all $n \in \mathbb{N}$[6].

(c) For all $\delta > 0$, $\displaystyle\lim_{n\to\infty} \int_{-\infty}^{-\delta} |f_n(t)|\mathrm{d}t = 0$ and $\displaystyle\lim_{n\to\infty} \int_{\delta}^{\infty} |f_n(t)|\mathrm{d}t = 0$.

Example. Let

$$f_n = \begin{cases} c_n(1-x^2)^n, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Choose each c_n such that $\displaystyle\int_{-\infty}^{\infty} f_n(t)\mathrm{d}t = \int_{-1}^{1} f_n(t)\mathrm{d}t = 1$. Then (a) and (b) are satisfied. We have $f_n(x) \geq \dfrac{1}{2}$ on $\left[-\dfrac{1}{2\sqrt{n}}, \dfrac{1}{2\sqrt{n}}\right]$ so

$$\int_{-\infty}^{\infty} (1-x^2)^n\mathrm{d}x = \int_{-1}^{1} (1-x^2)^n\mathrm{d}x \geq \int_{-\frac{1}{2\sqrt{n}}}^{\frac{1}{2\sqrt{n}}} (1-x^2)^n\mathrm{d}x \geq \int_{-\frac{1}{2\sqrt{n}}}^{\frac{1}{2\sqrt{n}}} \frac{1}{2}\mathrm{d}x = \frac{1}{2\sqrt{n}}$$

Thus $c_n \leq 2\sqrt{n}$. For $\delta > 0$ we have $c_n(1-x^2)^n \leq 2\sqrt{n}(1-\delta^2)^n$ on $(-\infty, -\delta) \cup (\delta, \infty)$. Thus

$$\int_{-\infty}^{\delta} |f_n(x)|\mathrm{d}x = \int_{-1}^{-\delta} |f_n(x)\mathrm{d}x \leq \int_{-1}^{-\delta} 2\sqrt{n}(1-\delta^2)^n\mathrm{d}x \leq 2\sqrt{n}(1-\delta^2)^n$$

[6] This is redundant if $f_n \geq 0$ for all $n \in NN$

which goes to 0 as $n \to \infty$. With the same holding for $\int_\delta^\infty |f_n(x)|\mathrm{d}x$. So (c) is satisfied and $\{f_n\}$ forms an approximate identity.

Theorem C. Let $\{f_n\}$ be an approximate identity. Let $g : \mathbb{R} \to \mathbb{C}$ be bounded and uniformly continuous. Then $f_n * g \to g$ uniformly on $\mathbb{R}$.

Proof. Select M_1 such that for all n,

$$\int_{-\infty}^{\infty} |f_n(x)|\mathrm{d}x \leq M_1$$

and M_2 such that

$$|g(x)| \leq M_2 \qquad (x \in \mathbb{R}).$$

Let $\varepsilon > 0$, then there exists δ such that for all $x, y \in \mathbb{R}$ with $d(x, y < \delta$,

$$|g(x) - g(y)| < \varepsilon.$$

For $x \in \mathbb{R}$, we have

$$\begin{aligned}
|f_n * g(x) - g(x)| &= \left| \int_{-\infty}^{\infty} f_n(t)g(x-t)\mathrm{d}t - \int_{-\infty}^{\infty} f_n(t)g(x)\mathrm{d}t \right| \\
&= \left| \int_{-\infty}^{\infty} f_n(t)(g(x-t) - g(x))\mathrm{d}t \right| \\
&\leq \int_{-\infty}^{\infty} |f_n(t)(g(x-t) - g(x))|\mathrm{d}t \\
&= \int_{-\infty}^{-\delta} |f_n(t)(g(x-t) - g(x))|\mathrm{d}t + \int_{-\delta}^{\delta} |f_n(t)(g(x-t) - g(x))|\mathrm{d}t \\
&\quad + \int_{\delta}^{\infty} |f_n(t)(g(x-t) - g(x))|\mathrm{d}t \\
&\leq 2M_2 \int_{-\infty}^{\delta} |f_n(t)|\mathrm{d}t + \varepsilon \int_{-\delta}^{\delta} |f_n(t)|\mathrm{d}t + 2M_2 \int_{\delta}^{\infty} |f_n(t)|\mathrm{d}t \\
&= 2M_2 \left(\int_{-\infty}^{\delta} |f_n(t)|\mathrm{d}t + \int_{\delta}^{\infty} |f_n(t)|\mathrm{d}t \right) + \varepsilon \int_{-\delta}^{\delta} |f_n(t)|\mathrm{d}t
\end{aligned}$$

We can choose N large enough such that for all $N \geq N$,

$$\int_{-\infty}^{\delta} |f_n(t)|\mathrm{d}t < \varepsilon \text{ and } \int_{\delta}^{\infty} |f_n(t)|\mathrm{d}t < \varepsilon.$$

So for $n \geq N$,

$$|f_n * g(x) - g(x)| \leq \varepsilon(4M_2 + M_1).$$

□

Idea of Proof. Split the integral into 3 parts which we can bound by assumptions of approximate identity.

Note. If g is not bounded, then we are not able to take g outside of the integral. And if g is not uniformly continuous, we can't take the ε outside of the middle integral.
An example violating the uniform continuity assumption is $\sin(1/x)$.

Definition. Let (X, d) be a metric space, and $f : X \to \mathbb{R}$ or $f : X \to \mathbb{C}$. We say f has compact support if there exists some compact set K such that $f(x) = 0$ for all $x \in X \backslash K$.
If $X = \mathbb{R}$, then $f : \mathbb{R} \to \mathbb{C}$ has compact support if and only if there exists R such that $f(x) = 0$ for all $x \notin [-R, R]$.

Lemma D. Let $f : \mathbb{R} \to \mathbb{C}$ or $f : \mathbb{R} \to \mathbb{R}$ be compact supported and integrable. Let $Q : \mathbb{R} \to \mathbb{C}$ be a polynomial. Then $f * Q$ is a polynomial.
If $f : \mathbb{R} \to \mathbb{R}$ and Q has real coefficients, then $f * Q$ has real coefficients.

Proof. Write

$$Q(x) = \sum_{k=0}^{n} a_k x^k$$

with $x^0 = 1$. Then

$$\begin{aligned}
(f * Q)(x) &= \int_{-\infty}^{\infty} f(t) Q(x-t) \mathrm{d}t \\
&= \int_{-R}^{R} f(t) Q(x-t) \mathrm{d}t \\
&= \int_{-R}^{R} f(t) \sum_{k=0}^{n} a_k (x-t)^k \mathrm{d}t \\
&= \int_{-R}^{R} f(t) \sum_{k=0}^{n} a_k \sum_{l=0}^{k} \binom{k}{l} (-t)^{k-l} x^l \mathrm{d}t \\
&= \int_{-R}^{R} \sum_{k=0}^{n} \sum_{l=0}^{k} f(t) a_k \binom{k}{l} (-t)^{k-l} x^l \mathrm{d}t \\
&= \sum_{k=0}^{n} \sum_{l=0}^{k} \left(\int_{-R}^{R} f(t) a_k \binom{k}{l} (-t)^{k-l} \mathrm{d}t \right) x^l
\end{aligned}$$

□

Idea of Proof. Write out Q as a sum and use binomial theorem, then take sums outside of the integral.

Theorem 7.26 (Weierstrass approximation theorem). Let $f : [a, b] \to \mathbb{C}$ or $f : [a, b] \to \mathbb{R}$ continuous, theen there exists a sequence of polynomials $\{P_n\}_{n \in \mathbb{N}}$ such that $P_n \to f$ uniformly on $[a, b]$.
If $f : [a, b] \to \mathbb{R}$, then $\{P_n\}$ can be chosen to have real coefficients.

Proof. Step 1: We want to reduce Theorem 7.26 to the case where $[a, b] = [0, 1]$ and $f(0) = 0 = f(1)$.
Suppose 7.26 is true for such functions and now let $g : [a, b] \to \mathbb{C}$ be continuous. Let

$$f_1(x) = g(a + x(b - a))$$

and

$$f_2(x) = f_1(x) - (f_1(0)(1 - x) + f(1)x).$$

Hence if $Q_n \to f_2$ uniformly, then

$$P_n(x) = Q_n\left(\frac{x-a}{b-a}\right) + \left(f_1(0)\left(1 - \frac{x-a}{b-a}\right) + f_1(1)\left(\frac{x-a}{b-a}\right)\right).$$

Hence it suffices to prove 7.26 for $f : [0, 1] \to \mathbb{C}$ with $f(0) = f(1) = 0$.
Step 2: Extend $f : \mathbb{R} \to \mathbb{C}$ by setting $f(x) = 0$ for $x \notin [0, 1]$. Then f this f is uniformly continuous and bounded. By Theorem C, if $\{\widetilde{Q_n}\}$ is an approximate identity then $\widetilde{Q_n} * f \to f$ uniformly. We will use $\widetilde{Q_n}$ from the example:

$$\widetilde{Q_n} = \begin{cases} c_n(1 - x^2)^n, & -1 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

and

$$Q^n = c_n(1 - x^2)^n$$

Step 3: Prove $\widetilde{Q_n} * f(x) = Q_n * f(x)$ for all $x \in [0, 1]$.
Let $x \in [0, 1]$. Then

$$\begin{aligned}
\widetilde{Q_n} * f(x) = f * \tilde{Q_n}(x) &= \int_{-\infty}^{\infty} f(t)\widetilde{Q_n}(x - t)\mathrm{d}t \\
&= \int_0^1 f(t)\widetilde{Q_n}(x - t)\mathrm{d}t \\
&= \int_0^1 f(t)Q_n(x - t)\mathrm{d}t \\
&= \int_{-\infty}^{\infty} f(t)Q_n(x - t)\mathrm{d}t \\
&= f * Q_n(x) \\
&= Q_n * f(x).
\end{aligned}$$

By Lemma D, $Q_n * f$ is a polynomial, so $\widetilde{Q_n} * f$ is as well. □

Idea of Proof. Prove if it works for easier functions on [0,1] it works fro all functions. Then extend the function to all of R. Then we know there's a polynomial that goes to that function.

Definition 7.28. (A) Let E be a set. Let $\mathcal{A}$ be a set of functions $f : E \to \mathbb{C}$ or $E \to \mathbb{R}$. We say $\mathcal{A}$ is a (complex/$\mathbb{C}$) algebra if for all $f, g \in \mathcal{A}$, $c \in \mathbb{C}$

(a) $f + g \in \mathcal{A}$.

(b) $f \cdot g \in \mathcal{A}$.

(c) $cf \in \mathcal{A}$.

(B) Let $\mathcal{A}$ be an algebra. We say $\mathcal{A}$ is uniformly closed if for all uniformly convergent sequences $\{f_n\} \subset \mathcal{A}$, we have

$$\lim_{n\to\infty} f_n \in \mathcal{A}.$$

(C) Let $\mathcal{A}$ be an algebra. Let $\mathcal{B} = \{f : E \to \mathbb{C} |$ there exists $\{f_n\} \subset \mathcal{A}$ s.t $f_n \to f$ uniformly$\}$. In other words $\mathcal{B}$ is the set of "limits of uniformly convergent sequences in $\mathcal{A}$". $\mathcal{B}$ is called the uniform closure of $\mathcal{A}$ and sometimes we write it is $\overline{\mathcal{A}}$.

Example. Here are some examples of algebras:

(I) The set of polynomials $f : \mathbb{R} \to \mathbb{C}$.

(II) $\mathcal{C}(\mathbb{R})/\mathcal{C}(\mathbb{C})$.

(III) Trigonometric polynomials

$$\sum_{k=0}^{n} (a_k \sin(kx) + b_k \cos(kx))$$

(IV) Symmetric polynomials[7]

(V) Piece-wise polynomial, continuous functions.

(VI) Functions of the form

$$\sum_{k=0}^{n} c_k e^{2\pi ikx} / \sum_{k=-n}^{n} c_k e^{2\pi ikx}$$

(VII) Holomorphic functions on $\mathbb{C}$.

We can then restate the Weierstrass approximation theorem:

Theorem 7.26 (Weierstrass approximation theorem). Let $E = [a, b]$, $\mathcal{A} = \{\text{polynomials}([a, b])\}$. Then $\overline{\mathcal{A}} = \mathcal{C}([a, b])$.

Note. If $\mathcal{A}$ is an algebra of bounded functions then it has a metric $d(f, g) = \sup_{x\in E} |f(x) - g(x)|$. So $(\mathcal{A}, d)$ is a metric space and is a subset of the larger metric space (X, d) where $X = \{$ bounded $f : E \to \mathbb{C}\}$. The uniform closure of $\mathcal{A}$ is the close of $\mathcal{A}$ in the metric space X.

Theorem 7.29. Let $\mathcal{A}$ be an algebra of bounded functions. Then $\overline{\mathcal{A}}$ is a uniformly closed algebra.

[7] `https://en.wikipedia.org/wiki/Symmetric_polynomial`

Proof. Recall (X, d) is the metric space of bounded functions ($f : E \to \mathbb{C}$ or $E \to \mathbb{R}$) and $\overline{\mathcal{A}}$ is the close of $\mathcal{A}$ in (X, d). Hence $\overline{\mathcal{A}}$ is closed. Let $f, g \in \overline{\mathcal{A}}$ and $c \in \mathbb{R}$ or $c \in \mathbb{C}$. Let $\{f_n\}$ be a sequence in $\mathcal{A}$ such that $f_n \to f$ uniformly, and $\{g_n\}$ such that $g_n \to g$ uniformly. Then it is easy to see that

$$\begin{aligned} f_n + g_n &\to f + g \text{ uniformly} \\ f_n \cdot g_n &\to f \cdot g \text{ uniformly} \\ cf_n &\to cf \text{ uniformly} \end{aligned}$$

□

Idea for Stone-Weierstrass. Let K be a compact set, $\mathcal{A}$ a (real) algebra of continuous functions $f : K \to \mathbb{R}$. Then $\overline{\mathcal{A}}$ is the algebra of all continuous functions $f : K \to \mathbb{R}$ unless there is an obstruction.

This begs the question. What is an obstruction to this theorem ? Some such obstructions are the 0-algebra or algebras of constant functions.

Definition 7.30. (A) Let E be a set of functions $f : E \to \mathbb{R}$ or $E \to \mathbb{C}$. We say $\mathcal{A}$ separates points if for all $x, y \in E$, there exists $f \in \mathcal{A}$ such that $f(x) \neq f(y)$.

Failure to separate points is an obstruction to Stone-Weierstrass.

Then even polynomials fail to separate points.

(B) Let E and $\mathcal{A}$ as before. We say that $\mathcal{A}$ vanishes at no point of E if for all $x \in E$, there exists $f \in \mathcal{A}$ such that $f(x) \neq 0$.

Vanishing at a point in E is an obstruction to Stone-Weierstrass.

Theorem 7.32 (Stone-Weierstrass Theorem). Let K be a compact metric space, $\mathcal{A}$ an algebra of continuous functions $f : K \to \mathbb{R}$. Suppose that $\mathcal{A}$ separates points and vanishes at no point. Then the uniform closure of K is there algebra of all continuous functions $f : K \to \mathbb{R}$.

Proof. The proof of this theorem has 5 main steps that we will write as lemmas.

Lemma 1. Let K and $\mathcal{A}$ as above. If $f \in \overline{\mathcal{A}}$, then $|f| \in \overline{\mathcal{A}}$.

Proof. Let $f \in \overline{\mathcal{A}}$, let $M = \sup_{x \in K} |f(x)|$. Then $M < \infty$. Let $\varepsilon > 0$. By Theorem 7.26, there exists a polynomial A with

$$|Q(y) - |y|| < \frac{\varepsilon}{2} \qquad (y \in [-M, M]).$$

Let $P(y) = Q(y) - Q(0)$. So $P(0) = 0$ and

$$|P(y) - |y|| = |Q(y) - Q(0) - |y| - |0|| \leq |Q(y) - |y|| + |Q(0) - |0|| < \varepsilon.$$

Write $P(x) = \sum_{k=1}^{n} c_k x^k$. Then $\mathcal{A}$ is an algebra so

$$P(f) = \sum_{k=1}^{n} c_k f^k = g \in \mathcal{A}.$$

For $x \in K$,

$$|g(x) - |f(x)|| = |P(f)(x) - |f(x)|| < \varepsilon.$$

□

Lemma 2. Let K and $\mathcal{A}$ as before. Let $f_1, \ldots, f_n \in \overline{\mathcal{A}}$. Then $\max(f_1, \ldots, f_n) \in \overline{\mathcal{A}}$ and $\min(f_1, \ldots, f_n) \in \overline{\mathcal{A}}$.

Proof. We will prove this by induction. Let $f_1, \ldots, f_n \in \overline{\mathcal{A}}$. Then

$$\max(f_1, \ldots, f_n) = \max(\max(f_1, \ldots, f_{n-1}), f_n) = \max(g, f).$$

Hence is suffices to show if $f, g \in \overline{\mathcal{A}}$, then $\max(g, f) \in \overline{\mathcal{A}}$.
Rewriting the max and min functions we have

$$\max(f, g) = \frac{1}{2}(f + g) + \frac{1}{2}|f - g| \in \overline{\mathcal{A}}$$

and

$$\min(f, g) = \frac{1}{2}(f + g) - \frac{1}{2}|f - g| \in \overline{\mathcal{A}}.$$

□

Lemma 3 (Theorem 7.31). Let K and $\mathcal{A}$ as above. Let $x, y \in K$ with $x \neq y, c, d \in \mathbb{R}$. Then there exists $f \in \mathcal{A}$ such that $f(x) = c$ and $f(y) = d$.

Proof. Since $\mathcal{A}$ separates points and vanishes at no point, there exists $g, h, k \in \mathcal{A}$ such that

$$g(x) \neq g(y), h(x) \neq 0 \text{ and } k(y) \neq 0.$$

Let

$$u(z) = g(z)k(z) - g(x)k(z) = (g(z) - g(x))k(z) \in \mathcal{A}$$

and

$$v(z) = (g(z) - g(y)h(z) \in \mathcal{A}.$$

Then $u(x) = 0, u(y) \neq 0$, $v(x) \neq 0$ and $v(y) = 0$. Set

$$f(z) = \frac{c}{v(x)} v(z) + \frac{d}{u(y)} u(z) \in \mathcal{A}.$$

Then $f(x) = c$ and $f(y) = d$. □

Lemma 4. Let K and $\mathcal{A}$ as before. Let $f : K \to \mathbb{R}$ be continuous, $c \in K$ and $\varepsilon > 0$. Then there exists $g \in \overline{\mathcal{A}}$ such that

$$g(x) = f(x) \text{ and } g(t) - f(t) \geq -\varepsilon \qquad (t \in K).$$

Proof. For each $y \in K$, use Lemma 3 to find $g_y \in \mathcal{A}$ such that $g_y(x) = f(x)$, $g_y(y) = f(y)$. Since $g_y \in \mathcal{A}$, g_y is continuous so $g_y - f$ is continuous and

$$g_y(y) - f(y) = 0.$$

Hence there exists an open set $U_y \subset K$ such that

$$g_y(t) - f(t) \geq -\varepsilon \qquad (t \in U_y).$$

Then $y \in U_y$ and K is compact so there exists a finite sub-cover $U_{y_1}, \ldots U_{y_n}$ of K.
Let $g = \max(g_{y_1}, \ldots, g_{y_n}) \in \overline{\mathcal{A}}$ by Lemma 2.
Then

$$g(x) = \max(g_{y_1}(x), \ldots, g_{y_n}) = \max(f(x), \ldots, f(x)) = f(x).$$

For $t \in K$, t is contained in some U_{y_j}, so

$$g(t) \geq g_{y_j}(t)$$

which gives

$$g(t) - f(t) \geq g_{y_j}(t) - f(t) \geq -\varepsilon.$$

□

Lemma 5. Let K and $\mathcal{A}$ as before. Let $f : K \to \mathbb{R}$. Then there exists $h \in \overline{\mathcal{A}}$ such that

$$|h(t) - f(t)| < \varepsilon \qquad (t \in K).$$

Proof. For $x \in K$, let $g_x \in \overline{\mathcal{A}}$ by Lemma 4. The n$g_x - f$ is continuous and $g_x(x) - f(x) = 0$. Hence there exists an open set U_x containing x such that

$$g_x(t) - f(t) < \varepsilon \qquad (t \in U_x).$$

Since K is compact, we can select a finite sub-cover $U_{x_1}, \ldots, U_{x_n}$. Let

$$h = \min(g_{x_1}, \ldots, g_{x_n}) \in \overline{\mathcal{A}}.$$

By the same argument as Lemma 4,

$$h(t) - f(t) < \varepsilon \qquad (t \in K).$$

And so

$$h(t) - f(t) \geq -\varepsilon \qquad (t \in K).$$

Hence $|h(t) - f(t)| < \varepsilon$. □

To conclude the proof, let $f \in \mathcal{C}_{\mathbb{R}}(K)$. For all n, select $g_n \in \overline{\mathcal{A}}$ such that

$$||f - g|| < \frac{1}{2n}.$$

Select $f_n \in \mathcal{A}$ such that

$$||f_n - g_n|| < \frac{\varepsilon}{2}.$$

Then

$$||f - f_n|| \leq ||f - g_n|| + ||g_n - f_n|| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $f_n \to f$ uniformly. □

We would like to be able to make a similar statement, replacing $\mathcal{C}_{\mathbb{R}}(K)$ with $\mathbb{C}(K)$, however this would false.

Example. Let S' be the unit circle in $\mathbb{C}$. Then $f : S' \to \mathbb{C}$ can be represented as $f(z)$ with $z \in S$ or $f(e^{it})$ with $t \in \mathbb{R}$ and $f(e^{i0}) = f(e^{i2\pi})$.
Let $g(z) = \overline{z} \in \mathcal{C}(K)$.
If $\mathcal{A}$ is the space of complex valued polynomials,

$$\int_0^{2\pi} P(e^{it})e^{it}\mathrm{d}t = 0$$

because $P(z)$ is a holomorphic function and we are integrating along a closed contour. However,

$$\int_0^{2\pi} P(e^{it})e^{it}\mathrm{d}t = 2\pi$$

So if there is a sequence $P_n \to g$ uniformly, then $P_n z \to g_n z$ uniformly and

$$0 = \int_0^{2\pi} P_n(e^{it})e^{it}\mathrm{d}t \to \int_0^{2\pi} P(e^{it})e^{it}\mathrm{d}t = 2\pi$$

giving a contradiction. Hence $\overline{z} \notin \overline{\mathcal{A}}$.

Definition. Let $\mathcal{A}$ be a complex algebra. We say $\mathcal{A}$ is self adjoint if for every $f \in \mathcal{A}$, $\overline{f} \in \mathcal{A}$.

Theorem 7.33 (Complex Stone-Weierstrass). Let K be a compact metric space and $\mathcal{A}\mathcal{C}(K)$ where $\mathcal{A}$ is an algebra who separates points, vanishes at no point and is self-adjoint. Then $\overline{\mathcal{A}} = \mathcal{C}(K)$.

Proof. Let $\mathcal{A}_{\mathbb{R}}$ be the (real) algebra of real-valued functions in $\mathcal{A}$.
Step 1: Let $f = u + iv \in \mathcal{A}$, then

$$u = \frac{1}{2}(f + \overline{f}) \in \mathcal{A} \text{ hence } \mathrm{Re}(f) \in \mathcal{A}_{\mathbb{R}}$$

and

$$v = \frac{1}{2}(f - \overline{f}) \in \mathcal{A} \text{ hence } \mathrm{Im}(f) \in \mathcal{A}_{\mathbb{R}}.$$

Step 2: $\mathcal{A}_{\mathbb{R}}$ separates points.
Let $x, y \in K$ with $x \neq y$. By Lemma 3[8], there exists $f \in \mathcal{A}$ such that $f(x) = 0$ and $f(y) = 1$. So $\mathrm{Re}(f)(x) = 0$, $\mathrm{Re}(f)(y) = 1$. Hence $\mathcal{A}_{\mathbb{R}}$ separates points.

Step 3: $\mathcal{A}_{\mathbb{R}}$ vanishes at no point.
Let $x \in K$. Since $\mathcal{A}$ vanishes at no point, there exists $f \in \mathcal{A}$ such that $f(x) \neq 0$. Then at least one of $\mathrm{Re}(f)(x) \neq 0$ or $\mathrm{Im}(f)(x) \neq 0$ must be true.[9]

Step 4: By Stone-Weierstrass, $\overline{\mathcal{A}_{\mathbb{R}}} = \mathcal{C}_{\mathbb{R}}(K)$.

Step 5: Let $f = u + iv \in \mathcal{C}(\mathcal{K})$ then $u, v \in \mathcal{C}_{\mathbb{R}}(K)$. So there exists $g, h \in \mathcal{A}_{\mathbb{R}}$ such that

$$||u - g|| < \frac{\varepsilon}{2} \text{ and } ||v - h|| < \frac{\varepsilon}{2}.$$

Hence

$$||f - (g + ih)|| \leq ||u - g| + ||iv - ih|| < \varepsilon.$$

$\square$

[8]Lemma 3 only uses that $\mathcal{A}$ is an algebra that separates points and vanishes at no point. In fact Lemma 3 holds for any field, not just $\mathbb{R}$.

[9]Two other arguments hold: $f(x) = re^{i\theta} \Rightarrow e^{-i\theta} f(x) = r$ and $f\overline{f} = |f|^2$.

Chapter 8

Fourier Series

8.1 Vector Spaces

Definition. Denote the field $\mathbb{F}$ to be one of $\mathbb{Q}, \mathbb{R}, \mathbb{C}$.

A **vector space** V over $\mathbb{F}$ is a non-empty set on which two operations (**addition** and **scalar multiplication**) are defined so that for each pair of elements x, y in V there is a unique element $x + y$ in V and for each element a in $\mathbb{F}$ and each element x in V there is a unique element ax in V such that the following conditions hold.

(VS 1) For all x, y in V, $x + y = y + x$ (commutativity of addition).

(VS 2) For all x, y, z in V, $(x + y) + z = x + (y + z)$ (associativity of addition).

(VS 3) There exists an element in V denoted 0 such that $0 + x = x$ for each x in V.

(VS 4) For each element x in V there is an element y in V such that $x + y = 0$.

(VS 5) For each element x in V, $1x = x$.

(VS 6) For each pair of element a, b in $\mathbb{F}$ and each element x in V, $(ab)x = a(bx)$.

(VS 7) For each element a in $\mathbb{F}$ and each pair of elements x, y in V, $a(x + y) = ax + ay$.

(VS 8) For each pair of elements a, b in $\mathbb{F}$ and each element x in V, $(a + b)x = ax + bx$.

The elements $x+y$ and ax are called the **sum** of x and y and **product** of a and x, respectively.

Example. (I) $\mathbb{R}^d, \mathbb{C}^d$.

(II) Sequences $(x_n)_{n\in\mathbb{N} \text{ or } n\in\mathbb{Z}}$ with $x_i \in \mathbb{R}$ or $x_i \in \mathbb{C}$.

(III) Sequences $(x_n)_{n\in\mathbb{N} \text{ or } n\in\mathbb{Z}}$ where $x_n = 0$ for all $n \geq N$.[1]

[1] The is the space of all sequences which converge to 0 with respect to the identity metric.

(IV) Sequences $(x_n)_{n\in\mathbb{N} \text{ or } n\in\mathbb{Z}}$ where $\sum |x_n| < \infty$.

(V) Sequences $(x_n)_{n\in\mathbb{N} \text{ or } n\in\mathbb{Z}}$ where $\left(\sum |x_n|^p\right)^{\frac{1}{p}} < \infty$, $1 \leq p < \infty$. This is also know as the vector space ℓ^p.

(VI) $\mathcal{R}_\alpha[a, b]$.

Definition. A norm is a function $||\cdot|| : V \to \mathbb{R}$ that satisfies the following properties for all:

(i) $||x + y|| \leq ||x|| + ||y||$.

(ii) $||ax|| = |a| \cdot ||x||$.

(iii) $||x|| \geq 0$ and $||x|| = 0$ if and only if $x = 0$.

Where $x, y \in V, c \in \mathbb{F}$.

Example. (I) $V = \mathbb{R}^d$ or $\mathbb{C}^d$ $||x||_p = \left(\sum_{j=1}^{n} |x_j|^p\right)^{\frac{1}{p}}$. When $p = 2$ is the euclidean norm.

(II) $V = \ell^p$ and $||a||_p = \left(\sum |a_n|^p\right)^{\frac{1}{p}}$

(III) $V = \mathcal{C}([0, 1])$, $||f|| = \int_0^1 |f(t)|\mathrm{d}t$

(IV) $V = \mathcal{C}([0, 1])$, $||f||_{L^p} = \left(\int_0^1 |f(t)|^p \mathrm{d}t\right)^{\frac{1}{p}}$. This is know as the L^p norm.

(V) $V = \mathcal{C}([0, 1])$, $||f||_{L^\infty} = \sup_{x\in[0,1]} |f(x)|$. This is know as the L^∞ norm.

(VI) $V = \mathcal{R}[0, 1]$ with the L^p norm is not a norm because the third property fails. We can take $f(x) = \begin{cases} 0, & x \in [0, 1) \\ 1, & x = 1 \end{cases}$ then $\left(\int_0^1 |f(t)|^p \mathrm{d}t\right)^{\frac{1}{p}} = 0$.

We can fix this by defining the equivalence relation $\sim$ on $\mathcal{R}[0, 1]$ with $f \sim g$ if $\int_0^1 |f(t) - g(t)|\mathrm{d}t = 0$.

Let $V = \{[f] : f \in \mathcal{R}[0, 1]\} = \mathcal{R}[0, 1]/\sim$. Then we can define

$$||[f]||_p = \left(\int_0^1 |f(t)|^p \mathrm{d}t\right)^{\frac{1}{p}}.$$

Definition. A pair $(V, ||\cdot||)$ is called a normed vector space. This induces he metric $d(x, y) = ||x - y||$.
If (V, d) is complete, then we call $(V, ||.||)$ a Banach space.

Example. $(\mathbb{R}^d, ||\cdot||_p)$, is a Banach space but $(\mathcal{C}([0,1]), ||\cdot||_{L^p})$ is not a Banach space. However for L^∞ it is a Banach space.

Definition. Let V be a vector space, $\langle , \rangle$ an inner product. Then $(V, \langle , \rangle)$ is called an inner product space. This inner product induces a norm $||x|| = (\langle x, x\rangle)^{\frac{1}{2}}$ which in turn induces a metric d.
If (V, d) is complete, then we call $(V, \langle , \rangle)$ a Hilbert space.

Example. (I) $V = \ell^2(\mathbb{N})$ with $\langle a, b\rangle = \sum a_n \overline{b_n}$.

(II) $V = \mathcal{C}([0,1])$ with $\langle f, g\rangle = \displaystyle\int_0^1 f(t)\overline{g(t)}\mathrm{d}t$.

(III) $V = \mathcal{C}([-\pi,\pi])$ with $\langle f, g\rangle = \displaystyle\int_{-\pi}^{\pi} f(t)\overline{g(t)}\mathrm{d}t$

The inner product space we are most interested in for this course is $V = \mathcal{R}([0,1])/\sim$ with the inner product

$$\langle [f], [g]\rangle = \int_0^1 f(t)\overline{g(t)}\mathrm{d}t$$

Instead of the above we will simply write

$$\langle f, g\rangle = \int_0^1 f(t)\overline{g(t)}\mathrm{d}t.$$

Note. Let $\{f_n\}, f : [0,1] \to \mathbb{C}$. If $[f_n] \to [f]$ in $L^2([0,1])$, i.e.

$$\int_0^1 |f_n(x) - f(x)|^2 \mathrm{d}x \to 0$$

as $n \to \infty$, does $f_n \to f$ pointwise ? The answer is no. Since $[f] = [g]$ does not imply $f = g$, we can have $f_n \to g$ for any $g \in [f]$.
Consider $f(x) = \dfrac{1}{x}$ and $f_n(x) = f(x)\xi_{[\frac{1}{n},1]}$. Is $\{f_n\}$ Cauchy in $L^2([0,1])$? No because the gap between f_n and f_m is too large. However with $f = \dfrac{1}{x^3}$ this is fine.
Here is an example of a sequence which converges $L^2([0,1])$ but $f_n \not\to f$ pointwise:

$$f_n(x) = \begin{cases} 1, & \text{for } x \in [h_n, h_{n+1}] \quad \text{mod } h_n = \displaystyle\sum_{i=1}^{n} \frac{1}{i} \\ 0, & \text{elsewhere} \end{cases}$$

Then

$$\left(\int_0^1 |f_n - f_m|^2\right)^{\frac{1}{2}} \leq \left(\int_0^1 |f_n|^2\right)^{\frac{1}{2}} + \left(\int_0^1 |f_m|^2\right)^{\frac{1}{2}} = \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}.$$

8.2 Fourier series

Definition 8.10. If $\{\phi_n\}$ is sequence of Riemann integrable functions on $[a,b]$ that are normal, i.e.

$$\langle \phi_n, \phi_m \rangle = \int_a^b \phi_n(x)\overline{\phi_m(x)}\mathrm{d}x = \begin{cases} 1, & \text{if} n = m \\ 0, & \text{if} n \neq m \end{cases}$$

then we call $\{\phi_n\}$ a "system of orthonormal functions" or an "orthonormal system" on $[a,b]$.

Example. $\{e^{2\pi inx}\}$, $\{\sin(n\pi x)\} \cup \{\cos(n\pi x)\}$ and the wavelet basis are all orthonormal systems.

Definition. If $\{\phi_n\}$ is an orthonormal system on $[a,b]$ and if $c_n = \langle f, \phi_n \rangle$, then we call c_n the n-th Fourier coefficient of f relative to $\{\phi_n\}$.
If $[a,b] = [0,1]$ and $\phi_n = e^{2\pi inx}$ we write $c_n = \hat{f}(n)$.

Definition. If $\{c_n\}$ are the Fourier coefficients of f relative to some orthonormal system $\{\phi_n\}$, then $\sum_{n=1}^{\infty} c_n\phi_n$ is called the Fourier series of f and we write $f \sim \sum_{n=1}^{\infty} c_n\phi_n$.

Note. This says nothing about the convergence of the RHS. We can also notice that the Fourier series is the projection of f onto the orthonormal system.

Theorem 8.11. Let $\{\phi_n\}$ be an orthonormal system on $[a,b]$ and let $f \in \mathcal{R}_\alpha[a,b]$ with Fourier coefficients $\{c_n\}$. Let

$$s_n = \sum_{j=1}^{n} c_j\phi_j(x)$$

by the n-th partial Fourier sum of f. Let $d_1, \ldots, d_n \in \mathbb{C}$ and

$$t_n = \sum_{j=1}^{n} d_j\phi_j(x).$$

Then $||f - s_n|| \leq ||f - t_n||$ with equality if and only if $s_n = t_n$, i.e. $d_j = c_j$ for all $j = 1, 2, \ldots, n$.

Proof. Let us first compute

$$\langle f, t_n \rangle = \int_a^b f(x)\overline{t_n(x)}\mathrm{d}x = \int_a^b f(x)\sum_{j=1}^{n}\overline{d_j\phi_j}(x)\mathrm{d}x = \int_a^b \sum_{j=1}^{n}\overline{d_j}f(x)\overline{\phi_j}(x)\mathrm{d}x = \sum_{j=1}^{n}\overline{d_j}c_j.$$

Similarly, $\langle t_n, f \rangle = \sum_{j=1}^{n}\overline{c_j}d_j$. Then

$$||t_n||^2 = \int_a^b t_n\overline{t_n}\mathrm{d}x = \int_a^b \left(\sum_{j=1}^{n} d_j\phi_j(x)\right)\left(\sum_{j=1}^{n}\overline{d_j\phi_j}(x)\right)\mathrm{d}x = \sum_{j=1}^{n} d_j\overline{d_j} = \sum_{j=1}^{n}|d_j|^2.$$

Now we compute

$$\begin{aligned}
||f-t_n||^2 = \int_a^b |f-t_n|^2 &= \int (f-t_n)\overline{(f-t_n)} \\
&= \int |f|^2 - \int f\overline{t_n} - \int \overline{f}t_n + \int |t_n|^2 \\
&= \int |f|^2 - \sum_{j=1}^n \overline{d_j}c_j - \sum_{j=1}^n \overline{c_j}d_j + \sum_{j=1}^n |d_j|^2 \\
&= \int |f|^2 - \sum_{j=1}^n |c_j|^2 + \sum_{j=1}^n |d_j - c_j|^2 \\
&\geq \int |f|^2 - \sum_{j=1}^n |c_j|^2.
\end{aligned}$$

If $d_j = c_j$ for all j, then

$$||f-t_n|| = ||f||^2 - ||s_n||^2. \tag{1}$$

So

$$||f-t_n||^2 = \int |f-t_n|^2 \geq ||f||^2 - ||s_n||^2 = ||f-s_n||^2$$

with equality if and only if $s_n = t_n$. □

Note. As a consequence of (1) we know

$$\sum_{j=1}^n |c_j|^2 - ||s_n||^2 \leq ||f||^2. \tag{2}$$

hence

$$\sum_{j=1}^n |c_j|^2 \leq ||f||^2.$$

This is known as the Bessel inequality.
If $f \in$ span$\{\phi_n\}$ then we have equality.

Definition. Let $L > 0$. We say $f : \mathbb{R} \to \mathbb{C}$ is L-periodic if

$$f(x+L) = f(x) \qquad (x \in \mathbb{R}).$$

Example. $e_n(x)$ is 1-periodic and, $\sin(x), \cos(x)$ are 2π-periodic.

From now on we will set $[a,b] = [0,1]$ and $\{\phi_n\}_{n\in\mathbb{Z}}$ where

$$\phi_n(x) = e^{2\pi inx} = e_n(x),$$

and the vector space

$$V = \{f : \mathbb{R} \to \mathbb{C} | f \text{ is 1-periodic } \& \ f \in \mathcal{R}([0,1])/\sim\}.$$

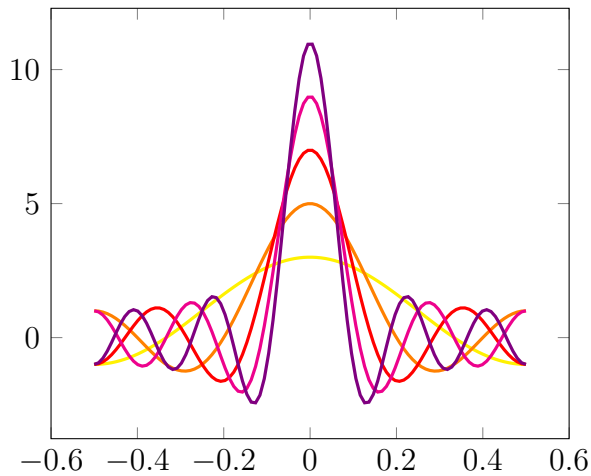


Figure 8.1: $D_N(t)$ for $N = 1, 2, 3, 4$ on $[-0.5, 0.5]$.

On V we define the inner product

$$\langle f, g\rangle = \int_0^1 f(x)\overline{g(x)}\mathrm{d}x.$$

Note. We have $e_n : G \to \mathbb{C}$ but since $|e_n(x)| = 1$, $e_n : G \to \{c \in \mathbb{C} : |c| = 1\}$. Furthermore $e_n e_m = e_{(n+m)}$, that is e_n is an operation over the group $\mathbb{Z}$.

Definition. The N-th partial Fourier series of f is

$$s_N(x) = \sum_{n=-N}^{N} c_n e_n(x).$$

Lemma 8.13 (Summation formula for Fourier series). For $x \in \mathbb{R}$,

$$s_N(x) = \int_0^1 f(x-t)D_N(t)\mathrm{d}t$$

where

$$D_N(t) = \sum_{n=-N}^{N} e^{2\pi int} = \frac{\sin\left(2\pi\left(N+\frac{1}{2}\right)t\right)}{\sin(\pi t)}$$

is called the Dirichlet kernel.

Proof. Step 1:

$$D_N(t) = e^{2\pi int}\sum_{k=0}^{2N} e^{2\pi ikt} = e^{-2\pi iNt}\frac{e^{2\pi i(2N+1)t}-1}{e^{2\pi it}-1}\left(\frac{e^{-\pi it}}{e^{-\pi it}}\right) = \frac{e^{2\pi i(N+0.5)t}-e^{-2\pi i(2N+1)t}}{e^{\pi it}-e^{-\pi it}}$$
$$= \frac{\sin\left(2\pi\left(N+\frac{1}{2}\right)t\right)}{\sin(\pi t)}$$

Step 2:

$$\begin{aligned}
s_N(X) &= \sum_{n=-N}^{N} c_n e^{2\pi i n x} = \sum_{n=-N}^{N} \langle f_n, e_n \rangle e^{2\pi i n x} \\
&= \sum_{n=-N}^{N} \int_0^1 f(t) e^{2\pi i n(-t)} \mathrm{d}t e^{2\pi i n x} \\
&= \sum_{n=-N}^{N} \int_0^1 f(t) e^{2\pi i n(x-t)} \mathrm{d}t \\
&= \int_0^1 f(t) D_N(x-t) \mathrm{d}t \\
&= \int_x^{x+1} f(x-s) D_N(s) \mathrm{d}s \\
&= \int_0^1 f(x-s) D_N(s) \mathrm{d}s
\end{aligned}$$

□

Definition. Let $f : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$. We say f is Lipschitz continuous at x if there exists $\delta > 0$ and $L > 0$ such that for all $y \in \mathbb{R}$ with $|x - y| < \delta$ we have

$$|f(x) - f(y)| \leq L|x - y|.$$

It is clear that if f is Lipschitz continuous, it is Lipschitz continuous at every point but the converse is not true.

Theorem 8.14. Let $f : \mathbb{R} \to \mathbb{C}$ be 1-periodic and $f \in \mathcal{R}([0,1])$. Suppose f is Lipschitz continuous and some point x. Then

$$\lim_{N \to \infty} S_N(x) = f(x).$$

Proof. Let's begin with some useful computations.

(1) $\displaystyle\int_0^1 D_N(t) \mathrm{d}t = \langle D_N, 1 \rangle = 1.$

(2) $\displaystyle D_N(t) = \frac{\sin(\pi t)\cos(2\pi N t) + \sin(2\pi N t)\cos(\pi t)}{\sin(\pi t)} = \cos(2\pi N t) + \cos(\pi t)\sin(2\pi N t).$

(3) If g is 1-periodic and $g \in \mathcal{R}([0,1])$

$$\sum_{n=-\infty}^{\infty} |c_n|^2 \leq \int_0^1 |g|^2 < \infty.$$

This is the Riemann-Lebesgue Lemma. Hence c_n $to 0$ and $|n| \to \infty$.

As a consequence, $\lim_{n\to\infty} \langle g, \sin(2\pi nx)\rangle \to 0$ because

$$|\langle g, \sin(2\pi nx)\rangle| = |\langle g, \frac{1}{2i}(e^{2\pi inx} - e^{-2\pi inx})\rangle| = \frac{1}{2}(|\langle g, e^{2\pi inx}\rangle + \langle g, e^{-2\pi inx}\rangle) = \frac{1}{2}(|c_n| + |c_{-n}|)$$

which goes to 0 as $n \to \infty$ and similarly, $|\langle g, \sin(2\pi nx)\rangle| \to 0$. Then,

$$\begin{aligned}
f(x) - s_N(x) &= f(x)\int_0^1 D_N(t)\mathrm{d}t - \int_0^1 f(x-t)D_N(t)\mathrm{d}t \\
&= \int_0^1 (f(x) - f(x-t))D_N(t)\mathrm{d}t \\
&= \int_0^1 (f(x) - f(x-t))\cos(2\pi Nt)\mathrm{d}t \\
&\quad + \int_0^1 (f(x) - f(x-t))\cot(\pi t)\sin(2\pi Nt)\mathrm{d}t \\
&= \langle f(x) - f(x-t), \cos(2\pi Nt)\rangle + \langle f(x) - f(x-t)\cot(\pi t), \sin(2\pi Nt)\rangle \\
&= \langle f(x) - f(x-t), \cos(2\pi Nt)\rangle + \langle h(t), \sin(2\pi Nt)\rangle
\end{aligned}$$

We know $\langle f(x) - f(x-t), \cos(2\pi Nt)\rangle \to 0$ as $N \to \infty$.
Now we want to show $h \in \mathcal{R}([0,1])$. Clearly h is periodic, so we need to show it is bounded. Since f is Lipschitz continuous at x there exists δ such that if $t \in [-\delta, \delta]$

$$|h(t)| = |(f(x) - f(x-t))\frac{\cos(\pi t)}{\sin(\pi t)}| \le M|t||\frac{\cos(\pi t)}{\sin(\pi t)}| = M|\cos(\pi t)||\frac{t}{\sin \pi t}| \le^* M$$

* We know $\sin(\pi t) \ge \frac{1}{2}\pi t$ for $t \in [-1/2, 1/2]$, so $\frac{t}{\sin(\pi t)} \le \frac{2}{\pi} \le 1$.
For $t \in [-1/2, 1/2]\backslash[-\delta, \delta]$

$$|h(t)| \le 4\sup_{z\in\mathbb{R}} |f(z)|\frac{1}{\pi\delta}.$$

So h is bounded and is Riemann integrable on $[-1/2, 1/2]$. Hence $\langle h(t), \sin(2\pi Nt)\rangle \to 0$ as $N \to \infty$. □

As a consequence we have the two following corollaries:

Corollaries.

(i) If $f(x) = 0$ for all x in an open interval I, then $s_N(x) \to 0$ for all $x \in I$.

(ii) If $f(x) = g(x)$ on an interval I, then $s_N(f, x) - s_N(g, x) \to 0$ for all $x \in I$.

Theorem 8.15. Let $f : \mathbb{R} \to \mathbb{C}$ be 1-period and continuous. Let $\varepsilon > 0$. Then there exists a trigonometric polynomial

$$P = \sum_{n=-N}^{N} a_n e^{2\pi inx}$$

such that

$$\sup_{x\in\mathbb{R}} |f(x) - P(x)| < \varepsilon.$$

Proof. We want to apply Stone-Weierstrass. Here our metric space is $S' = \{z \in \mathbb{C} : |z| = 1\}$. Define $F : S' \to \mathbb{C}$ such that

$$F(e^{2\pi i t} = f(t)$$

This is well define since $f(t)$ is 1-periodic. Then let $\mathcal{A}$ be the algebra of trigonometric polynomials. This is self adjoint because

$$\overline{\sum_{n=-N}^{N} a_n z^n} = \sum_{n=-N}^{N} \overline{a_n} z^{-n} = \sum_{n=-N}^{N} a_{-n} z^n \in \mathcal{A}.$$

□

Note. (i) The trigonometric polynomial found in 8.15 is not necessarily $s_N(f,t)$.

(ii) There are some continuous, 1-periodic functions where $s_N(f)$ does not converge pointwise. So there exists continuous 1-periodic functions such that $s_N(f) \to f$ pointwise but not uniformly. However if

$$\int_0^1 |f|^p < \infty$$

then $s_N(f) \to f$. This is the Carlson-Hunt theorem.

(iii) In Rudin, problem 8.15 describes a sequence of trigonometric polynomials which converge to f uniformly

$$\sigma_N = \frac{s_0(f) + s_1(f) + s_2(f) + \cdots + s_N(f)}{N+1}.$$

This is the Cesàro mean.

Given the shortcomings of s_N, we can ask why do we use it them ? Well them have some nice properties compared to other approximations.

Theorem (Plancherel Theorem/Parseval-Plancherel Theorem). Let $f : \mathbb{R} \to \mathbb{C}$ be 1-periodic and integrable on [0,1]. Then

$$\lim_{N\to\infty} ||f - s_N(f)||_2 = 0.$$

i.e

$$\lim_{N\to\infty} \left(\int_0^1 |f(x) - s_N(f,x)|^2 \mathrm{d}x \right)^{\frac{1}{2}} = 0$$

or $s_N(f) \to f$ in $(L^2([0,1], ||\cdot||_2)$.

Proof. Let's recall Minkowski's inequality

$$||f+g||_2 \leq ||f||_2 + ||g||_2.$$

For $f \in \mathcal{R}([0,1])$ and $\varepsilon > 0$, there exists $g : [0,1] \to \mathbb{C}$ continuous such that

$$||f - g||_2 < \varepsilon.$$

Let $\varepsilon > 0$. Select some continuous functions g such that

$$||f - g|| < \frac{\varepsilon}{3}.$$

Then we have

$$||s_N(f) - f||_2 \leq ||s_N(f) - s_N(g)|| + ||s_N(g) - g||_2 + ||g - f||_2.$$

Then

$$||s_N(f) - s_N(g)|| = ||s_N(f - g)||_2 \leq ||f - g||_2 < \frac{\varepsilon}{3}.$$

By Theorem 8.15, there exists a trigonometric polynomial P of degree N_0 such that

$$||g - P||_\infty < \frac{\varepsilon}{3}$$

which implies

$$||g - P||_2 < \frac{\varepsilon}{3}.$$

Hence by Theorem 8.11, if $N > N_0$,

$$||s_N(g) - g||_2 \leq ||P - g||_2 < \frac{\varepsilon}{3}.$$

□

Theorem (Cauchy Schwarz). There are several version to this theorem

(v1): In $\ell^2(\mathbb{Z})$:

$$\sum_{k=1}^{n} |a_k \overline{b_k}| \leq \left(\sum |a_k|^2\right)^{\frac{1}{2}} \left(\sum |b_k|^2\right)^{\frac{1}{2}}$$

i.e.

$$|\langle a, b\rangle| \leq \langle a, a\rangle^{\frac{1}{2}} \langle b, b\rangle^{\frac{1}{2}}$$

(v2): In $L^2([0, 1])$:

$$\int 0^1 |f(x)\overline{g(x)}| \mathrm{d}x \leq \left(\int_0^1 |f(x)|^2 \mathrm{d}x\right)^{\frac{1}{2}} \left(\int_0^1 |g(x)|^2 \mathrm{d}x\right)^{\frac{1}{2}}$$

i.e.

$$|\langle f, g\rangle| \leq ||f||_2 ||g||_2$$

This is a special case of Hölder's inequality (proven in Rudin exercise 6.10):

$$|\langle f, g\rangle| \leq ||f||_p ||g||_q$$

for all $1 \leq p \leq q < \infty$ with $\dfrac{1}{p} + \dfrac{1}{q} = 1$.

Theorem 8.16 (Parseval's Theorem). Let $f, g : \mathbb{R} \to \mathbb{C}$ be 1-periodic and integrable on $[0,1]$. Let $\{c_n\}, \{d_n\}$ be their Fourier coefficients.[2] Then

$$\int_0^1 f(x)\overline{g(x)}\mathrm{d}x = \sum_{n\in\mathbb{Z}} c_n\overline{d_n}.$$

In inner product notation,

$$\langle f, g\rangle = \langle \hat{f}, \hat{g}\rangle.$$

Proof. Let us compute

$$\langle s_n(f), g\rangle = \int_0^1 \sum_{n=-N}^{N} c_n e^{-2\pi nx}\overline{g(x)}\mathrm{d}x = \sum_{n=-N}^{N} c_n \overline{\int_0^1 e^{-2\pi inx} g(x)\mathrm{d}x} = \sum_{n=-N}^{N} c_n\overline{d_n}$$

i.e.

$$\left|\langle f, g\rangle - \sum_{n=-N}^{N} c_n\overline{d_n}\right| = |\langle f, g\rangle - \langle s_n, g\rangle| = |\langle f - s_n, g\rangle| \leq ||f - s_n||_2||g||_2.$$

We know this goes to 0 as $N \to \infty$. However, we need absolute converge to finish the proof. If the series $\sum_{n=-N}^{N} c_n\overline{d_n}$ is not absolutely convergent, the value of the series can change based on the order we take the sum.
By Cauchy-Schwarz on ℓ^2,

$$\sum_{n=-N}^{N} |c_n\overline{d_n}| \leq \left(\sum_{n=-N}^{N} |c_n|^2\right)^{\frac{1}{2}} \left(\sum_{n=-N}^{N} |d_n|^2\right)^{\frac{1}{2}} \leq ||f||_2||g||_2.$$

So $\sum_{n=-N}^{N} c_n\overline{d_n}$ converges absolutely. □

8.3 Completion of $L^2([0,1])$

Define $\mathcal{F} : L^2([0,1]) \to \ell^2([0,1])$ by

$$f \to \hat{f}.$$

We would like $||f||_2 = \langle f, f\rangle^{\frac{1}{2}} = \langle \hat{f}, \hat{f}\rangle^{\frac{1}{2}} = ||\hat{f}||_2$.
$\mathcal{F}$ is not onto, but the its image is dense in $\ell^2(\mathbb{Z})$ i.e. if $(c_n) \in \ell^2(\mathbb{Z})$,

$$f_N(x) = \sum_{n=-N}^{N} c_N e^{2\pi inx}$$

[2] An alternate notation is $\hat{f}(n) = c_n, \hat{g}(n) = d_n$

so $||f_N||_2 = \sum_{n=-N}^{N} |c_n|^2$ and so

$$\left(\sum_{n=-N}^{N} |f_N(x) - c_n|^2\right)^{\frac{1}{2}} = \left(\sum_{n=-N}^{N} |c_n|^2\right)\frac{1}{2}$$

which goes to 0 as $N \to \infty$

Theorem 8.17. Let (X_1, d_1), (X_2, d_2) be a metric space. Suppose (X_2, d_2) is complete. Let $g : X_1 \to X_2$ be distance preserving i.e. $d_1(x, x') = d_2(g(x), g(x'))$. Suppose that the image of g is dense. Then g extends to an isometry $\tilde{g}$ from the completion $(\tilde{X}_1, \tilde{d}_1)$ of (X_1, d_1) to (X_2, d_2).

Note. Recall that if (X_1, d_1) is a metric space, and $(\tilde{X}_1, \tilde{d}_1)$ is its completion, then

$$\tilde{X}_1 = \{\text{Cauchy sequences in } (X_1, d)\}/\sim$$

where $(x_n) \sim (x'_n)$ if $\lim_{n\to\infty} d_1(x_n, x'_n) = 0$, and

$$\tilde{d}_1([(x_n)], [(x'_n)]) = \lim_{n\to\infty} d_1(x_n, x'_n) = \limsup_{n\to\infty} d_1(x_n, x'_n).$$

So

$$L^2[\tilde{0}, 1]$$

is the set of equivalence classes of Cauchy sequences of equivalence classes of Riemann integrable functions.

Proof. Define $\tilde{g} : (\tilde{X}_1, \tilde{d}_1) \to (X_2, d_2)$ as follows. Let $[(x_n)] \in X_1$. Then (x_n) is a Cauchy sequence so $(g(x_n))$ is a Cauchy sequence. Hence this converges to some $y \in X_2$.
Define $\tilde{g}([(x_n)]) = y$.
Let $\varepsilon > 0$ and $[(x_n)] = [(x'_n)]$. Then there exists N such that for all $n > N$, $d_1(x_n, x'_n) < \varepsilon$. Then since g is distance preserving, $d_2(g(x_n), g(x'_n)) < \varepsilon$. Hence $\lim_{n\to\infty} d_2(g(x_n), g(x'_n)) = 0$ and $\lim_{n\to\infty} g(x_n) = \lim_{n\to\infty} g(x'_n))$.
Then we need to check that $\tilde{g}$ is distance preserving, i.e.

$$\lim_{n\to\infty} \tilde{d}_1([(x_n)], [(x'_n)]) = d_2(\lim_{n\to\infty} g(x_n), \lim_{n\to\infty} g(x'_n)).$$

Then it suffices to show $\tilde{g}$ is onto.[3] Let $y \in X_2$. Since $g(X_1)$ is dense in X_2, there exists some sequence $(x_n) \in X_1$ such that $g(x_n) \to y$.
In particular, $(g(x_n))$ is Cauchy so (x_n) is Cauchy because g is distance preserving. Hence we can verify that $\tilde{g}([(x_n)]) = y$. □

Moral. Let $\tilde{L}^2[0, 1]$ be the completion of $L^2([0, 1])$ (THIS IS NOT STANDARD NOTATION).
Then $\mathcal{F}$ is an isometry from $\tilde{L}^2[0, 1] \to \ell^2(\mathbb{Z})$. Furthermore, $\langle f, g\rangle = \langle \hat{f}, \hat{g}\rangle$ is morally true.
In Math 420, $\tilde{L}^2[0, 1]$ is defined to be equivalence classes of Lebesgue integrable functions $f : [0, 1] \to \mathbb{C}$ with $\int_0^1 |f| < \infty$.

[3] It is trivially one-to-one because it is distance preserving.

Appendix A

Wikipedia Articles

- https://fr.wikipedia.org/wiki/Walter_Rudin
- https://fr.wikipedia.org/wiki/Michel_Rolle
- https://fr.wikipedia.org/wiki/Brook_Taylor
- https://fr.wikipedia.org/wiki/Bernhard_Riemann
- https://fr.wikipedia.org/wiki/Thomas_Joannes_Stieltjes
- https://fr.wikipedia.org/wiki/Frigyes_Riesz
- https://fr.wikipedia.org/wiki/Oliver_Heaviside
- https://fr.wikipedia.org/wiki/Augustin_Louis_Cauchy
- https://fr.wikipedia.org/wiki/Karl_Weierstrass
- https://fr.wikipedia.org/wiki/Ulisse_Dini
- https://fr.wikipedia.org/wiki/Archim%C3%A8de
- https://fr.wikipedia.org/wiki/Georg_Cantor
- https://fr.wikipedia.org/wiki/Cesare_Arzel%C3%A0
- https://fr.wikipedia.org/wiki/Giulio_Ascoli
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- https://fr.wikipedia.org/wiki/Stefan_Banach
- https://fr.wikipedia.org/wiki/David_Hilbert
- https://fr.wikipedia.org/wiki/Joseph_Fourier

- https://fr.wikipedia.org/wiki/Friedrich_Wilhelm_Bessel
- https://fr.wikipedia.org/wiki/Johann_Peter_Gustav_Lejeune_Dirichlet
- https://fr.wikipedia.org/wiki/Rudolf_Lipschitz
- https://fr.wikipedia.org/wiki/Henri-L%C3%A9on_Lebesgue
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- https://fr.wikipedia.org/wiki/Richard_Allen_Hunt
- https://fr.wikipedia.org/wiki/Ernesto_Ces%C3%A0ro
- https://fr.wikipedia.org/wiki/Michel_Plancherel
- https://fr.wikipedia.org/wiki/Marc-Antoine_Parseval_des_Ch%C3%AAnes
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